

ORIGINAL RESEARCH

Twist1 Promotes Endothelial Phenotypic Transition and Unstable Plaque Phenotype During Atherosclerosis

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BACKGROUND: Comprehensive investigation of endothelial cell (EC) dysfunction during atherosclerosis with single-cell omics has resulted in the proposal that ECs undergo multiple alternative cell fate decisions during disease, but lack of lineage tracing or spatial localization complicates interpretation of these data. *Twist1* , a causal gene for multiple atherosclerotic vascular diseases, is activated with low shear stress in ECs, and EC- *Twist1* knockout results in reduced atherosclerosis. However, it remains unclear how *Twist1* affects EC phenotype and plaque biology.

METHODS: We performed EC lineage tracing, in situ analysis, and scRNA-Seq in *ApoE^{-/-}* mice, both before disease and after 16 weeks of high-fat diet. We also performed these studies with 2 mouse models of EC *Twist1* deletion. We overexpressed *TWIST1* in human coronary artery ECs exposed to different flow conditions, followed by bulk RNA-seq. Human scRNA-Seq data were used to validate key findings in the mouse model.

RESULTS: We found that EC phenotypic modulation during atherosclerosis is characterized by both proinflammatory and endothelial-to-mesenchymal transition gene programs, occurring simultaneously along a single-cell fate transition. Human scRNA-Seq data validated a similar endothelial-to-mesenchymal transition during disease. We found that the commonly used *Twist1* conditional allele is hypomorphic, leading to reduced *Twist1* expression in multiple cell types. Using a mouse model of EC-specific *Twist1* deletion, we found reduced EC phenotypic modulation, decreased lesion size, and a more stable lesion phenotype. Integration of *TWIST1* overexpression in human coronary artery ECs with the mouse scRNA-seq data identified specific *TWIST1* -induced targets including CXCL12 and E-selectin during EC phenotypic modulation.

CONCLUSIONS: Our study revealed important aspects of EC phenotypic modulation during atherosclerosis, unifying disparate observations in the field. We identified key cellular and molecular mechanisms underlying a top risk locus for multiple vascular diseases, highlighting the promotion of inflammatory endothelial-to-mesenchymal transition by *TWIST1* as a key driver of disease risk.

Key Words: atherosclerosis ■ diet ■ endothelial cells ■ myocytes, smooth muscle ■ phenotype

Endothelial cell (EC) dysfunction, characterized by increased EC permeability, recruitment of circulating leukocytes, and increased thrombogenicity, is an important process that drives the progression of atherosclerosis and risk for myocardial infarction. These phenomena have been investigated extensively in vitro, but studies in vivo during atherosclerosis have typically relied on a small set of marker genes to describe EC phenotype. More recently, single-cell multiomic studies have allowed

more comprehensive phenotyping of multiple cell types, including ECs, during atherosclerosis. For example, a recent study revealed the potential for EC transitions to hematopoietic cells (endothelial-to-hematopoietic transition [EndHT]), immune-like cells (endothelial-to-immune cell-like transition [EndICLT]), stem/progenitor-like cells, and mesenchymal cells (endothelial-to-mesenchymal transition [EndMT]) phenotypes under disturbed flow and high-fat diet (HD)-induced atherosclerosis.¹ However,

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Nonstandard Abbreviations and Acronyms

DAPI	4',6-diamidino-2-phenylindole
DEG	differentially expressed gene
EC	endothelial cell
Edn1	endothelin 1
EndHT	endothelial-to-hematopoietic transition
EndICLT	endothelial-to-immune cell–like transition
EndMT	endothelial-to-mesenchymal transition
EV	empty vector
FGF-2	fibroblast growth factor-2
GFP	green fluorescent protein
HCAEC	human coronary artery endothelial cell
HFD	high-fat diet
IEL	internal elastic lamina
LSS	laminar shear stress
ModEC	modulated endothelial cell
ORO	Oil Red O
OSS	oscillatory shear stress
SELE	E-selectin
SMC	smooth muscle cell
TGF	transforming growth factor
UNC-CH	University of North Carolina at Chapel Hill

What Are the Clinical Implications?

Cardiovascular disease remains the leading cause of morbidity and mortality worldwide. Although current therapies primarily target risk factors, a deeper understanding of the biological processes driving atherosclerotic plaque formation is needed to develop more precise treatments. Endothelial cells line the inner surface of blood vessels, playing a central role in maintaining vascular homeostasis, and are among the first cells to respond to disease-promoting stimuli. Their activation initiates inflammatory signaling and phenotypic changes within the vessel wall that contribute to plaque development and plaque instability. We find that *TWIST1*, a major genetic risk factor for several vascular diseases, promotes inflammatory endothelial-to-mesenchymal transition, a process in which normally protective endothelial cells adopt a more inflammatory and fibrotic disease-promoting state. We also demonstrate that these mechanisms are conserved between mouse models and human vascular tissue, supporting the translational relevance of our findings. Deletion of *Twist1* leads to smaller plaques with fewer inflammatory cells and an overall more stable plaque phenotype. Importantly, we also identify key molecular signatures driven by *TWIST1* during disease. Together, these insights identify potential therapeutic targets aimed at preventing or reversing detrimental endothelial cell transitions, offering new strategies to limit the development and progression of cardiovascular disease.

these studies lacked either lineage tracing,^{1,2} comparison to nondiseased states,³ or spatial localization of cell populations identified in the single-cell data,^{2,3} leading to disparate views of this process due to the use of differing models. It is also unclear how potential EC populations identified using single-cell approaches correspond to the previously described aspects of EC dysfunction.

TWIST1, a basic helix-loop-helix transcription factor that regulates myriad processes during development, was found to be upregulated in the atheroprone, low shear stress, and lesser aortic curvature in the porcine aorta.⁴ Importantly, EC-specific *Twist1* knockout resulted in reduced lesion formation at an early atherosclerosis time point in a small number of mice.⁴ A more recent preprint report utilizing 8 weeks of atherosclerosis, followed by *Twist1* deletion and an additional 8 weeks of atherosclerosis, found reduced lesion size but a paradoxical increase in markers of plaque instability.⁵

We and others have identified *TWIST1* as the causal gene at the 7p21.1 locus associated at genome-wide significance with multiple atherosclerotic vascular diseases including coronary artery disease, ischemic stroke, and peripheral artery disease.⁶ Given the limited phenotyping of ECs during atherosclerosis in mice and humans, and the above studies suggesting an important role for *Twist1* in ECs during atherosclerosis, we sought to investigate the role of *Twist1* in ECs during atherosclerosis. The present study used (1) multiple mouse models for

EC lineage tracing, and EC-specific *Twist1* deletion in combination with detailed lesion phenotyping and single-cell transcriptomics in the setting of atherosclerosis; (2) integration of mouse and human data sets; and (3) bulk transcriptomics in human coronary artery ECs (HCAECs) exposed to *TWIST1* overexpression and differing flow conditions, to comprehensively define EC phenotypic modulation during atherosclerosis and the cellular and molecular mechanisms by which *TWIST1* modifies this process to influence lesion vulnerability and disease risk.

We found that EC phenotypic modulation is largely defined by a single phenotypic transition that encompasses both EndMT-like fibrotic changes and inflammatory EC phenotypes, but we did not find evidence of hematopoietic or immune cell transitions. Furthermore, we found that this EC phenotypic transition was similar in human samples. We show that *Twist1* deletion in ECs impairs this inflammatory EndMT transition in both the aortic root and arch, but only reduces atherosclerosis in the arch, suggesting additional factors are necessary for *Twist1*-induced EndMT to influence lesion formation. In the aortic arch, we found that EC-specific *Twist1* deletion reduced EndMT and resulted in reduced lesion size and a more stable lesion phenotype, consistent with the genetic evidence that *TWIST1* expression is associated

with increased vascular disease risk in humans, and in contrast to a recent preprint report of *Twist1* deletion in mice.⁵ Integration of (1) bulk RNA-seq of *TWIST1* overexpression in HCAECs, and (2) scRNA-seq of EC phenotypic modulation during atherosclerosis identified specific molecular mechanisms by which *TWIST1* modifies EC phenotype during disease, including upregulation of CXCL12 and SELE (E-selectin), as well as modification of EC junctions. The present study unites previous disparate findings regarding EC dysfunction and phenotypic modulation in vivo during atherosclerosis. It also demonstrates that a top genetic locus for multiple vascular diseases acts through *TWIST1* in ECs to promote an inflammatory EndMT phenotype resulting in lesion instability.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request. The single-cell RNA-sequencing data from mouse aorta and the bulk RNA-sequencing data from human coronary artery ECs generated in this study have been deposited in the NCBI Gene Expression Omnibus under accession number GSE342905. The Major Resources Table, which lists detailed information about the resources used in this study, can be found in the [Supplemental Material](#).

Statistical Methods

Analysis of Lesion Characteristics

Statistical analyses and data visualization were performed using GraphPad Prism (v9.5.0). For each quantitative comparison, data were first screened for outliers using the ROUT method ($Q=1\%$), and the remaining data were tested for normality using the Shapiro-Wilk test. For 2-group comparisons (Twist1-wt versus Twist1-ECKO, or Twist1-flx versus Twist1-ECKO), normally distributed data were compared using an unpaired, 2-tailed Student *t* test with Welch's correction (not assuming equal SDs), and non-normally distributed data were compared using an unpaired, 2-tailed Mann-Whitney *U* test. For comparisons across >2 groups (eg, Twist1-wt, Twist1-ECKO, and Twist1-flx aortic Twist1 expression), a 1-way ANOVA with Tukey post hoc test or a Kruskal-Wallis with Dunn post hoc test for multiple comparisons was used. Distributions of cells across pseudotime were compared using a 2-sample Kolmogorov-Smirnov test. Analyses were performed separately for male and female animals; data from the 2 sexes were not pooled. For all tests, a 2-sided $P<0.05$ was considered statistically significant. Exact *P* values are reported in the text and figure legends. Data are presented as mean $\pm$ SEM, with individual data points shown for all quantitative comparisons. The number of biological replicates (individual animals) or independent experiments for each comparison is stated in the corresponding figure legend.

Bulk RNA-Sequencing Differential Expression

Differential expression between TWIST1-overexpression and empty vector (EV; pWPI) HCAECs under each flow condition

was determined using DESeq2 (v1.28.0) on RSEM expected counts, applying an adjusted *P* value (Benjamini-Hochberg) cutoff of 0.05 and an absolute log2-fold-change cutoff of 1.

Single-Cell RNA-Sequencing Analysis

Differential gene expression between clusters and between experimental groups was performed in Seurat (v4.1.1) using the FindMarkers function with the default Wilcoxon rank-sum test. All differential expression testing was performed on the RNA assay (normalized counts). Genes were tested if detected in at least 10% of cells in either of the 2 populations being compared and if they exhibited a log fold-change of at least 0.25 between groups. *P* values were adjusted for multiple comparisons by Bonferroni correction based on the total number of genes in the data set, and an adjusted $P<0.05$ was considered statistically significant.

Mouse Strains

A tamoxifen-inducible Cre recombinase driven by the *Cdh5* promoter (Tg[Cdh5-cre/ERT2]1Rha) line was bred with a *tdTomato* fluorescent reporter line (B6.Cg Gt[ROSA]26Sortm14[AGtdTomato]Hze/J; 007914; JAX) to allow for the labeling of ECs and their progeny. A Twist1^{flx} allele was also bred onto these alleles. The Twist1^{flx} allele contains the following LoxP sites: 5' loxP site 472 bp upstream of Exon 1 and 3' loxP site within the noncoding Exon 2, located 201 bp upstream of the end of the exon. The loxP sites thus flank the *Twist1* promoter and entire coding sequence contained in exon 1. All alleles were on the C57Bl/6 *ApoE*^{-/-} background (B6.129P2-ApoE^{tm1Unc}/J). The final genotypes of the EC lineage tracing (Twist1-wt) mice were Tg^{Cdh5-CreERT2}, Twist1^{wt/wt}, ROSA^{tdT/+}, ApoE^{-/-}. The final genotypes of the lineage-traced Twist1 knockout (Twist1-ECKO) mice were: Tg^{Cdh5-CreERT2}, Twist1^{flx/flx}, ROSA^{tdT/+}, ApoE^{-/-}. Given our discovery that the Twist1^{flx} allele was hypomorphic in multiple cell types, we also generated an additional cohort of mice to ensure EC-specific deletion of *Twist1*. This additional cohort was all homozygous for the Twist1^{flx} allele, but included littermates that were both Cre+ and Cre-, thereby ensuring that any differences were the result of EC-specific *Twist1* deletion. Cre+ mice (still denoted Twist1-ECKO) had a final genotype of Tg^{Cdh5-CreERT2}, Twist1^{flx/flx}, ROSA^{tdT/+}, ApoE^{-/-}, and Cre- controls had a genotype of Twist1^{flx/flx}, ROSA^{tdT/+}, ApoE^{-/-}. Both male and female mice were included in this study. All mice were housed and bred in a pathogen-free facility at the University of North Carolina at Chapel Hill (UNC-CH), and experiments were conducted in accordance with the guidelines of UNC-CH's Institutional Animal Care and Use Committee. The animal study protocol was approved by the Administrative Panel on Laboratory Animal Care at UNC-CH.

Induction of Lineage Marker and Twist1 Deletion by Cre Recombinase

For all scRNA-Seq, RNAscope, and immunohistochemistry experiments, the tamoxifen gavage schedule was as follows: 2 doses of tamoxifen (T5648-56; Sigma-Aldrich), at 0.2 mg·g⁻¹ body weight, were administered by oral gavage at 6 to 7 weeks of age, with each dose separated by 48 hours. HFD (101511, Dyets; 21% anhydrous milk fat, 19% casein and 0.15% cholesterol) was started after the administration of both doses of tamoxifen, and animals remained on HFD for a total of 6

weeks or 16 weeks. A cohort of animals that were administered tamoxifen but did not receive HFD, deemed baseline, was included in the study. These animals were euthanized 72 hours after the last dose of tamoxifen.

Mouse Dissection and Perfusion

Mice were anesthetized using the open drop procedure with isoflurane and humanely euthanized by cervical dislocation. After sacrifice, a 22-gauge needle attached to a valve-controlled gravity perfusion system was inserted into the left ventricle, and 1× PBS was used to gently flush blood from the vasculature.

Mouse Aortic Root Dissociation for scRNA-Seq

For scRNA-Seq experiments, the aortic root and aortic arch (extending to just after the left subclavian artery) were excised. The tissue was washed 3× in ice-cold 1× PBS, then placed into 1 mL of an enzymatic dissociation cocktail (2 U/mL Liberase; 5401127001, Sigma-Aldrich) and 2 U/mL elastase (LS002279; Worthington Hank's Balanced Salt Solution) and minced with dissecting scissors. After incubation at 37°C for 30 minutes, the cells suspended in the enzyme cocktail were agitated by pipetting to ensure complete dissociation. The mixture was incubated at 37°C for an additional 30 minutes, resulting in a total incubation time of 1 hour. The cell suspension was then passed through a 70 µm strainer and centrifuged at 500g for 5 minutes at 4°C to pellet the cells. The enzyme solution supernatant was discarded, and cells were resuspended in fresh 1× Hank's Balanced Salt Solution and placed on ice until ready for sorting.

Fluorescence-Activated Cell Sorting of Mouse Aortic Root/Aortic Arch Cells

Immediately before sorting, the cell suspension was filtered through a 35 µm strainer. Cells were sorted on a benchtop SH800S cell sorter under UNC-CH's Advanced Analytics Core (supported by the Center for Gastrointestinal Biology and Disease Center Grant, P30 DK034987). Forward/side scatter parameters were used to exclude cell debris. Cells were then gated on forward scatter height versus forward scatter area to exclude doublet events. Cells that passed the aforementioned criteria were gated based on tdTomato fluorescence and sorted into either a 1.5 mL Eppendorf tube collecting lineage-traced ECs or a separate 1.5 mL Eppendorf tube collecting cells without detected fluorescence. Cells were then pooled to achieve enrichment for ECs (90% tdTomato-positive, 10% tdTomato-negative). ECs constitute a small population (≈1%–5%) of cells within the aortic tissue. Therefore, aortas from ≈10 mice were pooled per 10× capture to obtain sufficient ECs for downstream analysis. A total of 5 captures were performed: 1 baseline capture (wild-type mice ≈7 weeks of age on a chow diet), 2 captures from Twist1-wt mice after 16 weeks of HFD, and 2 captures from Twist1-ECKO mice after 16 weeks of HFD. Each pool included animals of both sexes, and sex was subsequently inferred at the single-cell level using *Ddx3y* and *Xist* expression. Cells of both sexes were distributed across all identified clusters (Figure S1A), supporting a sex-agnostic analytical approach for cell state identification.

Single-Cell Capture and Library Preparation

All single-cell capture and library preparation were performed at Advanced Analytics Core at the UNC-CH using the 10× Chromium platform according to the manufacturer's instructions. The Chromium Next GEM Single-Cell 3' Reagent Kit (v3.1 Dual Index) was used for library preparation (PN-1000268; 10X Genomics), and the sequencing platform used was the UNC High-Throughput Sequencing Facility NextSeq 2000 instrument (p3, flow cell). Data sets were later combined for all preceding analyses.

Mouse scRNA Data Analysis

FASTQ files from each experiment were aligned to a reference genome including the tdTomato transgene using Cell Ranger software version 7.1.0. Filtered matrix files were then processed using Seurat v4.1.1.⁸ Cells expressing fewer than 500 genes and genes expressed in <5 cells were removed from each data set. To identify potential doublets and poor-quality cells, the number of genes, unique molecular identifiers, and percentage of mitochondrial genes were examined. Further analysis of scRNA-Seq data was performed using Slingshot v2.2.1,⁹ Dynamo versus 1.3,¹⁰ and CellChat v1.0.0.¹¹

In total, 24854 high-quality cells were sequenced across Twist1-wt at baseline, Twist1-wt after 16 weeks of HFD, and Twist1-ECKO after 16 weeks of HFD data sets. Among these cells, 13262 were identified as tdTomato+/Cdh5+ ECs. Of these, 2856 aortic ECs underwent further analysis.

Cells expressing <500 genes and which had <5000 molecules detected within a cell were excluded from further analysis. Library size normalization was performed on gene expression values to normalize raw gene counts from each cell relative to the total number of read counts present in that cell. The values were multiplied by 10000 and log-transformed. Further analysis was performed using the 2000 most highly variable genes. Principal component analysis was used on the scaled data for dimensionality reduction, followed by clustering in principal component analysis space using a graph-based clustering approach. UMAP was used to visualize the resulting clusters in 2-dimensions. To analyze the ECs and endothelial-derived cells, only cells expressing the lineage-marker tdTomato above 1.5 were included. Considering that not all sequenced tdTomato+ cells are truly ECs or EC-derived, due to potential doublet events, the endothelial marker *Cdh5* was used to validate the identity of EC and EC-derived cells. To focus analysis on aortic ECs, an arterial EC module score was used, comprised of the following genes: *Gja5*, *Fbln5*, *Hey1*, *Ltbp4*, *Gja4*, and *Sox17*.^{12,13} Furthermore, we positively selected ECs expressing *Gja5*, *Cdh5*, and tdTomato (above a threshold of 1.5). We excluded cells based on expression of *Alpl* (microvascular marker), and also excluded lymphatic ECs (*Lyve1+Prox1+*), interstitial microvascular ECs (*Car4+*), and endocardial ECs and valve cells (*Tmem108+Prox1+*).¹⁴

Mouse and Human scRNA-seq Integration

Two human carotid data sets were used: one data set contained normal carotid samples derived from organ donors,¹⁵ and the other contained atherosclerotic vascular samples from patients undergoing carotid endarterectomy.¹⁶ ECs were isolated from these data sets using the markers *CDH5* and

GJA5, and microvascular ECs expressing *ALPL* were excluded. Human and mouse cells were integrated using the CONCORD algorithm (v1.0.13).¹⁷ Downstream analyses were performed using Seurat v4.1.1¹⁸ and scanPy v1.7.2.¹⁹

Mouse Aortic Root and Arch Preparation for Staining and Cryoblocks

Immediately after perfusion with 1× PBS, tissues being prepared for staining underwent a second perfusion with 0.4% ice-cold paraformaldehyde delivered by a separate valve-controlled gravity perfusion system. The aortic root was isolated separately from the remainder of the aorta, from the ascending aorta to just above the iliac arteries. The aortic root was left embedded in the myocardium of the upper half of the heart for on-axis mounting. Distal arch sections were isolated from the remainder of the arch. The tissues were then submerged in 4% paraformaldehyde overnight at 4°C, followed by immersion in a sucrose gradient up to 30% sucrose at 4°C. The tissue was embedded in OCT compound and frozen for cryoblock preparation.

En Face Oil Red O Staining and Quantification

Oil Red O (ORO) staining solution was prepared following the protocol by Chen et al²⁰ by dissolving ORO (A12989; Alfa Aesar) in isopropanol, followed by filtration with a 0.45 µm syringe filter (25-246; GenClone). The staining reagents were prepared fresh and used within 2 hours. Aortic arches were submerged in freshly filtered ORO for 1 hour at room temperature, then placed in 60% isopropanol for 20 minutes at room temperature and rinsed 3× for 5 minutes in double-distilled water. A stereomicroscope was then used to carefully trim away any perivascular adipose tissue and pin the aorta for en face imaging of the ORO stain. All ORO-stained areas were annotated using Fiji ImageJ, and ORO area was quantified as the %Area of the vessel.

Cryoblock Sectioning

Blocks were sectioned on a cryostat (Leica CM1950) into 7 µm thick sections, mounted on SuperFrost Plus Microscope Slides (1255015; Fisher Scientific) and kept at −80°C until slide preparation for staining, RNAscope, or IHC.

Fixation and Slide Mounting for tdTomato Signal

Slides were allowed to come to room temperature and air-dried before washing with 1X PBS to remove OCT. Slides were then placed in 95% ethanol for 15 seconds, air-dried, and immersed in cold 4% paraformaldehyde for 30 seconds for postfixation. Slides were then washed 4 times with deionized water before drying and mounting with a medium containing a DAPI (4',6-diamidino-2-phenylindole) stain (H-1200; Fisher Scientific).

Immunohistochemistry

Mouse aortic root slides were thawed at room temperature, and the OCT was removed by 2 washes of deionized water. Slides were submerged in 4% paraformaldehyde for 2 minutes,

followed by 4 washes in deionized water. A liquid-blocking pen was used to encircle sections of dried slides, followed by application of Peroxidized (PX968; Biocare Medical) for 5 minutes. Sections were washed 3× in deionized water before being incubated with Rodent Block M reagent (RBM961; Biocare Medical) for 30 minutes. Sections were washed twice in Tris-buffered saline (pH 7.4), then incubated overnight at 4°C with an anti-Cd68 rabbit polyclonal antibody (1:300 dilution; ab125212; Abcam), an anti-Cnn1 rabbit monoclonal (1:300 dilution; ab46794; Abcam), or a rabbit monoclonal isotype control (1:400 dilution; ab172730; Abcam). Sections were washed twice with Tris-buffered saline before being incubated with Rabbit-on-Rodent HRP Polymer (RMR622; Biocare Medical) for 30 minutes. Sections were washed twice with Tris-buffered saline and then incubated with the Betazoid DAB chromogen reagents (BDB2004; Biocare Medical) for 4 minutes. Sections were washed twice with deionized water and air-dried before mounting with Vectashield antifade mounting medium with DAPI (H-1200; Fisher Scientific).

RNAscope

Slides were washed once in PBS for 5 minutes to remove OCT, then baked at 60°C for 30 minutes. Slides were postfixed by immersion in 4% paraformaldehyde at 4°C for 15 minutes, followed by a dehydration series: 50% ethanol 5 minutes, 70% ethanol 5 minutes, 100% ethanol 5 minutes, 100% ethanol 5 minutes. Slides were air-dried, then treated with Hydrogen Peroxide (322335; ACD Bio) for 10 minutes. Slides were rinsed once in deionized water, then submerged in boiling 1× Target Retrieval reagent (322000; ACD Bio) for 5 minutes. Slides were washed twice in deionized water, then immersed in 100% ethanol and air-dried. Sections were encircled with a liquid-blocking pen and incubated with Protease Plus (322331; ACD Bio) reagent for 30 minutes at 40°C, then washed twice with deionized water. Sections were incubated with probes against mouse *Edn1* (endothelin 1; 435221; ACD Bio), *Alpl* (441161; ACD Bio), *Klk10* (584551; ACD Bio), or a negative control probe (320751; ACD Bio) for 2 hours at 40°C. Slides were processed according to the manufacturer's instructions, and all reagents were obtained from ACD Bio (322373). Sections were mounted with EcoMount (EM897L; Biocare Medical).

BODIPY Staining

After fixation, as previously described, each section was covered with a 35 µM solution of BODIPY 493 of 503 in PBS. The sections were incubated at RT for 30 minutes, protected from light, before washing 3× with PBS and mounting with VectaMount Express Mounting Medium (H-5700-60; Vector Laboratories).

Masson Trichrome Staining

Fixed slides were stained with Mayer's Hematoxylin Solution (MHS32-1L; Sigma-Aldrich) for 10 minutes, before washing with ddH₂O for 1 minute. Slides were then immersed in a Biebrich Scarlet-Acid Fuchsin solution (ICN1548552 and 857408; VWR) for 5 minutes, before washing with ddH₂O. Next, slides were differentiated in a phosphomolybdic-phosphotungstic acid solution (221856 and P4006;

Sigma-Aldrich) for 10 minutes before immediately being dipped into an aniline blue (AC22966205; Thermo Scientific) solution. Excess aniline blue was removed by washing slides in ddH₂O, followed by differentiation in a 1% acetic acid (A6283; Sigma-Aldrich) solution for 2 minutes. Slides were then washed with ddH₂O, followed by 95% EtOH, then 100% EtOH, and then cleared with xylene for 4 minutes. Slides were then mounted with DPX Slide Mounting Medium (06522; Sigma-Aldrich) and allowed to dry overnight before imaging.

Microscopy

Images were captured with the VS-200 SlideScanner at the UNC Hooker Imaging Core. Images for quantification of tdTomato and DAPI signal were captured with the fluorescent camera at 10X magnification. The normal Z-plane setting was selected for image capture, and the following channels at their respective exposure times were used: TRITC 50% (160.00 ms), fluorescein isothiocyanate 25% (75.70 ms), DAPI 50% (33.24 ms). Images for quantification of RNAscope-probed slides (*Edn1*) and for quantification of IHC-stained slides (CD68, Cnn1) were performed at ×20. Images were processed and quantified using the FIJI ImageJ software.²¹ Fluorescent images were quantified for tdTomato, and DAPI signal and brightfield images were quantified for RNAscope and IHC samples. Annotations of the lumen, internal elastic lamina (IEL), and external elastic lamina were outlined using the polygon selections tool. The endothelial monolayer was annotated as a 10-μm-wide segment²² adjoining the luminal surface. In each case, the software's enlarge selection function (Edit>Selection>Enlarge...) was used to create a duplicate of the lumen annotation with an input value of 10 μm, based on the image's scale. To ensure areas annotated by the automated enlarged annotation were not double-counted in cases where there was overlap with regions within the IEL, annotations of the inverse area of the IEL and its overlap with the lumen annotation were subtracted for final calculations of area and signal in the EC monolayer. These annotations are deemed regions of interest (ROIs) by the software. Two to 3 serial sections were analyzed for each aortic root sample for each feature quantified. The average of these serial sections was calculated to determine measurements per sample.

Vessel Feature Areas

The EC monolayer area was calculated by subtracting the lumen area from the area outlined 10 μm within the lumen annotation. Lesion area included the endothelial monolayer and was determined by subtracting the lumen area from the IEL area. Medial area was determined by subtracting the IEL area from the external elastic lamina area, and the vessel area was defined by the external elastic lamina area. Finally, an additional annotation that combined the inverse of the IEL and the EC monolayer was used to measure areas and signal areas overlapping between the EC monolayer and IEL, to prevent counting areas and signal areas twice. Researchers were blinded to the genotype of the animals until completion of the analysis.

Fluorescent Signal Measurement

All raw fluorescent images were split by channel and converted to an 8-bit image file. Each channel was processed

independently using the Li automated thresholding algorithm, which implements Li's Minimum Cross Entropy thresholding method²³ to binarize the image. Annotations were overlaid onto each channel, and each image was measured to obtain the area of each ROI and the %Area with signal for each ROI, based on the Li threshold.

Brightfield IHC Signal and RNAscope Signal Quantification

Following the image processing method outlined by Chen et al,²⁴ the background was subtracted from brightfield images of CD68 before further processing. To ensure only true IHC signal was included in quantification, color deconvolution was performed using the H DAB vector, and only the red Color 2 channel was further processed and quantified. Each image Look Up Table was inverted so that the white background would not be determined as signal area on applying the automated Li threshold. Measurements were recorded as described above. *Edn1* RNAscope signal was quantified by hand. Individual puncta were hand-counted using the multipoint tool. Puncta were quantified at the endothelial monolayer, within the lesion, and within the media separately. *Edn1* puncta are reported as an absolute count.

Calculations

The area of detected signal over a defined threshold (tdTomato and Cd68 signal) for each ROI was calculated by multiplying the area fraction (percent of ROI area with signal) by the area for the respective region of interest. The percentage of endothelial monolayer, lesion, media, and total vessel area with signal was determined by dividing the signal area by the total area for each respective region.

To determine the number of cells expressing a gene of interest (eg, tdTomato), the thresholded channels for each signal channel of interest (ex: DAPI channel and tdTomato channel) were combined using the (Process>Math>AND...) function to generate a composite image that only showed the colocalized signal area of the 2 channels. Using just the DAPI channel, 5 nuclei were selected at random for each of 3 images to be measured, and the average nuclei area was calculated. This calculated average was used as a standardized nucleus area; colocalized signal area was then divided by this standardized nucleus area to determine the number of cells expressing a gene of interest.

qPCR of Aorta for *Twist1* Expression

To validate the expression of *Twist1* in the aortic tissue collected from *Twist1*-wt, *Twist1*-ECKO, and *Twist1*-flx mice that did not receive tamoxifen, tissue was harvested and dissociated as previously described for scRNA-Seq sample preparation. For this validation, no fluorescence-activated cell sorting was performed, and all cell types were included. After centrifugation, instead of resuspending cells in 1× HBSS, cells from each mouse were resuspended in 250 μL of RNA Lysis Buffer from the Zymo Quick-RNA Microprep Kit (R1050, Zymo). RNA isolation was performed following the kit's standard protocol. cDNA synthesis was performed using the iScript cDNA Synthesis Kit (1708890, Bio-Rad). qPCR reactions were performed using the TaqMan Universal

MasterMix (4304437, Thermo Fisher), B2m VIC expression assay (4448489, Thermo Fisher), and *Twist1* FAM expression assay (4331182, Thermo Fisher) and run on a QuantStudio 6 Flex Real-Time Polymerase Chain Reaction System.

HiFi Cloning of *TWIST1* Into pWPI

The pWPI plasmid was obtained from Addgene (12254) and prepared using the ZymoPURE II Plasmid Midiprep Kit (D4201; Zymo Research). All plasmids were sequence-validated by GENEWIZ from Azenta Life Sciences. To clone *TWIST1* into the pWPI vector, the pWPI plasmid was digested with *Pme*I (R0560S; New England Biolabs) and 1X *r*Cut-Smart Buffer (B6004; New England Biolabs) at 37°C for 1 hour, then at 65°C for 20 minutes. A second incubation with QuickCIP (M0525S; New England Biolabs) in *r*Cut-Smart Buffer was performed at 37°C for 10 minutes, then at 80°C for 2 minutes. The Zymo DNA Clean & Concentrator-5 Kit (D4003; Zymo Research) was used on the suspended open backbone before continuing with assembly. Human *TWIST1* was obtained from Origene (NM_000474 Human Tagged ORF Clone, SKU RC202920; Origene) and cloned into the pWPI vector using NEBuilder High Fidelity (HiFi) DNA Assembly. After transformation into competent bacteria, plasmid DNA was purified with the ZymoPURE II Plasmid Midiprep Kit (D4201; Zymo Research). Primer sequences used for amplification of the *TWIST1* fragment with pWPI-compatible ends are as follows:

Forward primer: 5'-TATCGATCACGAGACTAGCCTCGAGGTTTGGAAACAAAAGTTGATTCTGAAGAAGATTT-3';

Reverse primer: 5'-CCGCCGCTTCCGCCGGAGCTGCCGCCTCCGGATCCGTGGGACGCGGACATGG-3'. For assembly of *TWIST1* into pWPI, a DNA molar ratio of 2:1 was used for *TWIST1* (insert) to pWPI (backbone) with 2× NEBuilder HiFi DNA Assembly Master Mix (E2621L; New England Biolabs). The reaction, in addition to an independent open vector-only negative control, was incubated at 50°C for 60 minutes, as instructed by the manufacturer's protocol.

Lentivirus Production

All lentivirus production was performed in a sterile cell culture hood. 293T cells (CRL-3216, ATCC; The Global Bioresource Center) were seeded into 100 mm Tissue Culture Dishes (Fisherbrand; FB012924) with 10 mL of DMEM (76176-654; VWR) with L-Glutamine (25-509; Genesee Scientific) and fetal bovine serum (F4135; Sigma-Aldrich). At 70% confluence, cells were transfected using the Invitrogen Lipofectamine 3000 Transfection Kit (L3000015; Invitrogen) protocol for pWPI and *TWIST1*-overexpression vectors. Transfected cells were incubated at 37°C in a cell culture CO₂ incubator for 6 hours, followed by a media change with 10 mL of warmed DMEM. Plates were incubated at 37°C in a cell culture CO₂ incubator for ≈62 hours before virus collection. GFP (green fluorescent protein) signal was detected after ≈24 hours of virus production. Lentivirus-containing media was aspirated with a 10 mL syringe and filtered through a 0.45 µm PES membrane (25-246; GenClone) and stored in 1 mL aliquots in cryovials (24-203; Genesee Scientific) at -80°C until ready for experimental use.

HCAEC Culture, Transduction, and Flow Model

HCAECs were ordered from Cell Applications (300-05a, Lot 2139) and cultured in Lonza's EBM-2 medium (CC-3156; Lonza) with EGM-2 Microvascular EC Growth Medium-2 BulletKit (CC-3202; Lonza), excluding gentamicin sulfate.

Once confluent in a T75 flask (25-209; GenClone), HCAECs were seeded for experiments at passage 4 into 6-well flat-bottom tissue culture plates (FB012927; Fisherbrand). At 80% confluence, each well of the 6-well plate containing HCAECs was incubated with 1 mL of EGM-2 microvascular medium, 1 mL of lentivirus, and 2 µL of Polybrene to increase transduction efficiency. For each condition (pWPI and *TWIST1*-overexpression), 4 biological replicates were prepared. After overnight incubation at 37°C in a cell culture CO₂ incubator, the medium was changed, with each well receiving 2 mL of fresh EGM-2 microvascular medium. Cells were incubated for 48 hours to allow for expression from the vectors; the GFP signal became apparent after 24 hours. After 48 hours, the HCAECs reached 100% confluence, and another medium change was performed before placing the cells on an orbital shaker (MS-NOR-30; Major Science), with 1.5 mL of medium per well. Plates were positioned at the corners of the shaking platform to ensure maximum orbital radius. HCAECs were exposed to 180 rpm for 5 days, and each day the 1.5 mL of EGM-2 microvascular medium was changed. After 5 days of shaking at 180 rpm, HCAECs were harvested for RNA extraction using the Zymo Quick-RNA MicroPrep Kit (11-327M; Zymo Research). For each well, a sterile cell scraper (12-565-58; Fisher Scientific) was used to detach laminar shear stress (LSS)-exposed HCAECs from the periphery of the well, and the cells suspended in medium were transferred to an Eppendorf tube and centrifuged at 500×*g* for 5 minutes at 4°C. The supernatant was aspirated and discarded, and the pellet of HCAECs was resuspended in 300 µL of RNA Lysis Buffer. After collecting LSS-exposed HCAECs from the periphery of each well, 1 mL of 1× PBS was added to each well to gently wash the well of uncollected detached cells. After aspirating the PBS, 300 µL of RNA Lysis Buffer was added to each well, and a scraper was used to detach the oscillatory shear stress (OSS)-exposed HCAECs at the center of each well. The suspended HCAECs were transferred to Eppendorf tubes, and RNA was extracted from all LSS-exposed and OSS-exposed HCAECs following the manufacturer's protocol, including an on-column DNase clean-up. RNA was suspended in DNase-free water and submitted to the Advanced Analytics Core at UNC for quality control and library preparation. RNA quantification and normalization were performed using the DeNovix RNA Fluorescence Assay, and RNA quality control was performed using the Agilent Bioanalyzer RNA Pico Kit. All RNA samples passed quality control with RIN scores >9. The mRNA-Seq library preparation was performed using the NEBNext Ultra II Directional Kit; library quantification and normalization were performed with the DeNovix DNA Fluorescence Assay, and library quality control was performed using the Agilent Bioanalyzer HS DNA Kit. Nf-SeqMatic Sequencing was performed by Admera.

Bulk RNA-Seq Data Processing and Analysis

FastQ files were processed using the nf-core/rnaseq (version 3.17.0)²⁵ pipeline.²⁶ Software environments from Bioconda²⁷ and Biocontainers²⁸ were used. Nextflow (v24.10.0) was used

to execute the pipeline.²⁹ Preprocessing to infer strandedness was performed using fq (v0.12.0) and Salmon (v1.10.3). The GRCh38.primary_assembly reference genome was used. Reads were trimmed using fastp (v0.23.4), and alignment was performed using STAR (v2.7.10a), and the count matrix data were generated using RSEM (v1.3.1) with a minimum mapped reads value of 5. Count matrix data were processed using DESeq2 (v1.28.0),³⁰ and expected counts data were used to determine differentially expressed genes (DEGs) between conditions. For all analyses, a *p*-adj cutoff of 0.05 and a log2FC cutoff of 1 were applied. DEG analysis was performed comparing *Twist1*-overexpression to pWPI EV under LSS conditions and OSS conditions. QIAGEN's Ingenuity Pathway Analysis software³¹ was used to determine the top predicted pathways and top predicted upstream regulators of DEGs.

RESULTS

Aortic EC Population Expands During Atherosclerosis and Contributes to the Plaque

To characterize EC phenotypic modulation during atherosclerosis, we generated an EC-specific lineage tracing mouse model (*Cdh5*^{CreERT2}, *ROSA26*^{tdT}, *ApoE*^{-/-}, or *Twist1*-wt). We administered tamoxifen to adult mice at 6 to 7 weeks of age before placing the animals on HFD for a course of 0, 6, or 16 weeks. At each respective time point, we isolated the aortic root for histological analysis. Fluorescent images of transverse sections of the aortic root were captured and annotated to define the different features of the vessel (Figure S1B). In situ analysis of the aortic root after 0, 6, and 16 weeks of HFD identified the presence of lineage-traced ECs at the EC monolayer of the aortic root, interfacing with the lumen. We also identified other ECs located outside of the vessel within the adventitia and myocardium, and lining the aortic valve leaflets (Figure 1A). We performed quantification of tdTomato and DAPI colocalization within the vessel (defined as the area between the lumen and the external elastic lamina) at each time point. This tdTomato-positive cell count revealed a significant increase in the absolute number of tdTomato-positive cells from 0 weeks to 16 weeks of HFD (Kruskal-Wallis with Dunn post hoc, *P*=0.004) and from 6 weeks to 16 weeks of HFD (Kruskal-Wallis with Dunn post hoc, *P*=0.047; Figure 1B), indicating the expansion of ECs in the aortic root. Within the atherosclerotic lesion (defined as the area between the EC monolayer and the IEL), the absolute number of tdTomato-positive cells significantly increased from 6 to 16 weeks of HFD (2-tailed Mann-Whitney *U* test *P*<0.001; Figure 1C). tdTomato-positive cells comprised 20% of all cells within the lesion at 6 weeks of HFD, and 15% of all cells in the lesion at 16 weeks of HFD. We found no significant difference in the number of tdTomato-positive cells within the 10 μ m-thick EC monolayer with disease progression (Figure S1C).

At the 0-week and 16-week time points, aortic tissue, extending from the aortic root to the distal edge of the

left subclavian artery takeoff, was isolated for enzymatic dissociation and fluorescence-activated cell sorting. We enriched for tdTomato-positive cells by pooling at a ratio of 90% tdTomato-positive cells to 10% tdTomato-negative cells, followed by scRNA-Seq capture using the 10x platform (Figure 1D). scRNA-Seq experiments included a total of 6 mice at 0 weeks of HFD (3 male, 3 female) and 14 mice at 16 weeks of HFD (9 male, 5 female).

scRNA-Seq analysis performed on the cells collected from the aortic root and arch revealed a diverse population, including various subpopulations (subclusters) of ECs (Figure 1E) expressing unique markers (Figure S1D and S1E). As the aortic root is partially embedded in the myocardium and also contains valvular tissue, we used cluster-specific markers to localize these EC populations within the aortic root. Nonaortic EC clusters were identified using previously described markers¹⁴ (Figure S1F through S1I) and excluded from further analyses. To positively identify arterial ECs lining the aortic lumen that participate in atherosclerosis in our mouse model, we used an arterial EC module score (Figure S1J), as well as coexpression of the well-established arterial EC marker *Gja5*,^{32,33} the general EC marker *Cdh5*,³⁴ and our lineage marker tdTomato (Figure 1F). A subcluster of these triple-positive cells, characterized by expression of the gene *Alpl*, was located exclusively in smaller vessels within the myocardium (microvascular ECs, Figure 1F; Figure S1K and S1L), so these were also excluded from further analyses. Thus, we focused on the population labeled aortic ECs in Figure 1F. Visualization of the data grouped by time point (0 weeks of HFD versus 16 weeks of HFD) showed that this aortic EC population was the only EC population that contained a disease-specific cell state (Figure 1G), further confirming that this group indeed represented the EC population participating in disease. Other disease-specific populations within the root included cells within the macrophage and smooth muscle cell (SMC) populations, which have been described previously.³⁵

On isolating the *tdT*⁺/*Cdh5*⁺/*Gja5*⁺/*Alpl*⁻ aortic ECs in UMAP space, we identified cells present at 0 weeks of HFD and deemed these baseline ECs. Some cells from the 16-week HFD (disease) time point were also present in this cluster, but many cells from the 16-week HFD time point formed a distinct disease-exclusive cluster comprised only of cells from that time point (Figure 1H). Thus, cells from the 16-week HFD disease time point appear to contain cells transiting from a quiescent to a phenotypically modulated EC (ModEC) phenotype found only in disease.

Among ECs present at the 16-week HFD disease time point, we analyzed DEGs between the phenotypically modulated ECs and the quiescent baseline ECs (Figure 1I). The heatmap featuring some of the most significant DEGs shows that genes involved in tight junctions (such as *Cldn5*³⁶ and *Cgln1*³⁷) and polarized

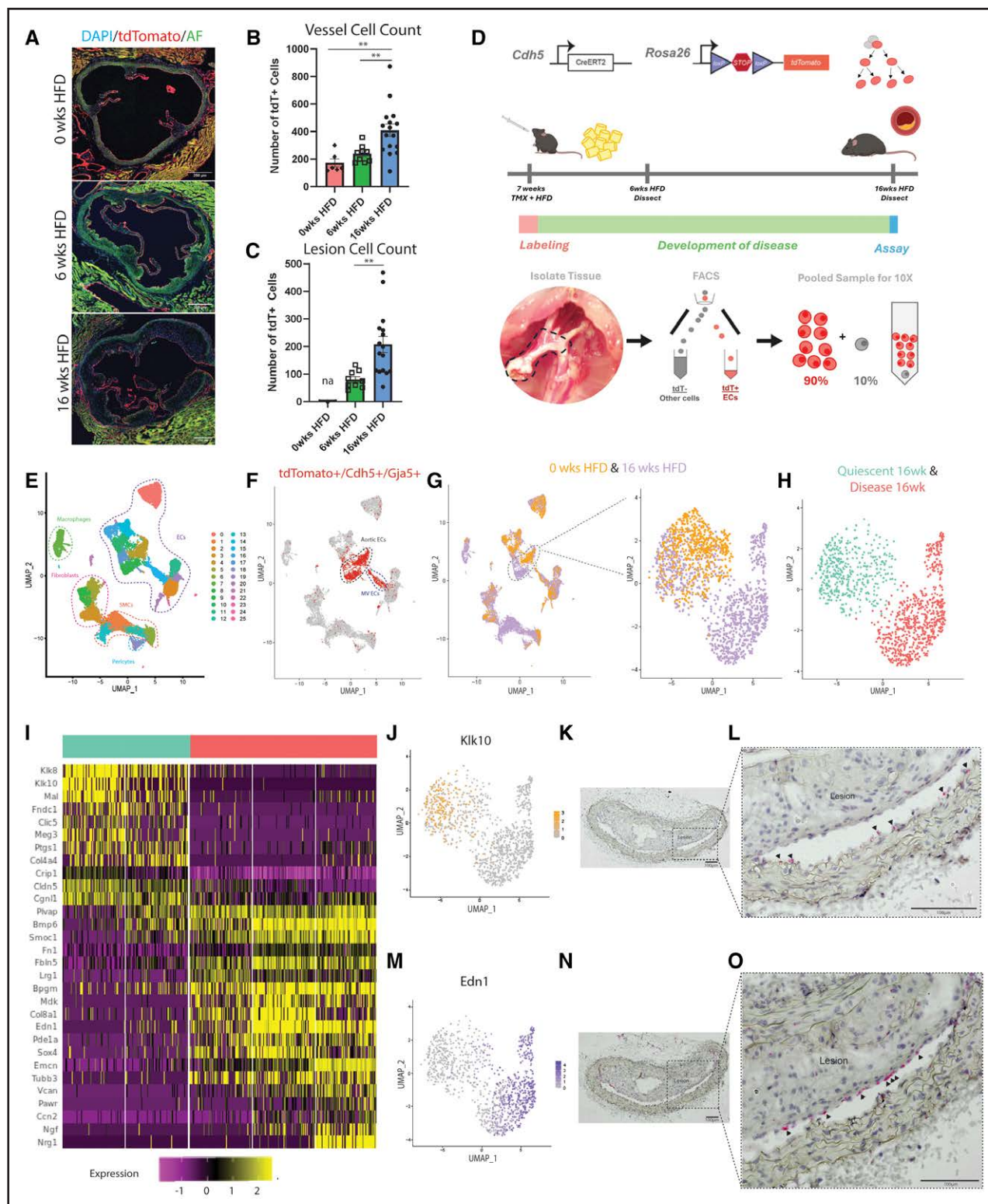


Figure 1. Identification of aortic luminal endothelial cells (ECs) and their phenotypic transition during disease.

A, Transverse sections of the aortic root after 0 (n=6), 6 (n=9), and 16 (n=16) weeks of high-fat diet (HFD), with nuclei represented in blue, lineage-traced tdTomato (tdT) cells represented in red, and tissue features represented by autofluorescence (AF) in green. **B**, Quantification of tdT+ cells, determined by tdT and DAPI (4',6-diamidino-2-phenylindole) colocalization count, in the vessel of male mice at each time point. Data were analyzed by Kruskal-Wallis with a Dunn post hoc test for multiple comparisons, and presented as mean±SEM. **C**, Quantification of tdT+ cells within the lesion of male mice at each time point. Data were analyzed by 2-tailed Mann-Whitney *U* test. Each point denotes the number of cells in 1 mouse. **D**, Schematic of lineage-traced Cre-lox system in a murine model of high-fat diet-induced atherosclerosis, outlining the experimental workflow of aortic arch isolation, fluorescence-activated cell sorting (FACS), and pooling of tdT+/tdT- cells for (Continued)

epithelial cells (such as *Klk8*, *Klk10*, and *Ma^β*) are highly expressed in the quiescent ECs but are lost in the ModEC clusters (Figure 1I). In contrast, genes such as *Plvap*, *Bmp6*, and *Edn1* show increased expression in the ModEC clusters, with *Ngf* and *Nrg1* being expressed in the most advanced ModEC cluster. Among the DEGs, we identified kallikrein-related peptidase 10, *Klk10*, as a specific marker of the quiescent EC cluster, as it was not expressed in the ModEC cluster (Figure 1J) nor in any other cell clusters in the scRNA-seq data (data not shown). RNAscope in situ hybridization of transverse sections of the mouse distal arch after 16 weeks of HFD shows the expression of *Klk10* at the quiescent EC monolayer (Figure 1K and 1L; Figure S1M). *Edn1* was identified as a specific marker of ModECs, as it was not highly expressed in the quiescent aortic EC cluster (Figure 1M; Figure S1N), nor in any other cell clusters in the data. RNAscope for *Edn1* in the distal arch sections after 16 weeks of HFD revealed expression of *Edn1* at the endothelial monolayer of the lesion and within the lesion below the monolayer (Figure 1N and 1O), confirming its ability to mark ECs undergoing phenotypic modulation, and also suggesting that ECs undergo phenotypic modulation at the monolayer before migrating into the lesion. *Nrg1*, a marker of advanced EC phenotypic modulation (Figure S1O), is also present at the EC monolayer and within the lesion (Figure S1P), although it is also present to a lesser extent within the valves and endocardium. The presence of *Nrg1* at the monolayer additionally suggests that the degree of EC phenotypic modulation does not necessarily correlate with EC migration into the lesion.

Single-Cell Analysis of EC Phenotypic Modulation Reveals a Single-Cell Fate Transition Defined by EndMT and Inflammatory Activation

Analysis of aortic ECs using the Slingshot package predicts a single trajectory beginning with the baseline ECs present at 0 weeks of HFD and continuing through the ModECs present at 16 weeks of HFD (Figure 2A), and RNA velocity analysis shows a similar cellular transition (Figure S2A). Slingshot pseudotime values along the trajectory are visualized in Figure 2B. As previous studies have identified various ECs phenotypes during atherosclerosis, we investigated known

markers of these phenotypes in our data. To investigate whether ECs adopt an EndMT phenotype, we visualized an EndMT Module Score (Figure 2C), comprised of well-established EndMT markers (*Acta2*, *Lum*, *Dcn*, *Fn1*, *Snail*, *Mmp2*, *Tagln*, *Tgfb1*, *Tgfb2*, *Tubb3*, and *Cdh2*). Selected components are shown in Figure 2D through 2F over Slingshot Pseudotime. The EndMT module score increased in the disease-exclusive ModECs (Figure 2D; Figure S2B), suggesting that these cells are undergoing or have undergone EndMT. We then assessed canonical endothelial markers (*Pecam1*, *Cdh5*, *Cd43*, *Cldn5*, *Klf2*, and *Vwf*), which were visualized both as a composite EC Module Score (Figure 2G) and individually (Figure 2H through 2J; Figure S2C) over Slingshot pseudotime using tradeSeq,³⁹ and observed an appreciable decrease in these markers. However, some canonical EC markers such as *Cdh5* were preserved even at advanced stages of EC phenotypic modulation (Figure S2D). Due to the persistence of *Cdh5* expression, it is possible that this represents a partial EndMT process.

Because many studies have described ECs acquiring a proinflammatory phenotype,⁴⁰ we also constructed a proinflammatory Module Score (Figure 2K), which included *Icam1*, *Selp*, *Sele*, *Cxcl12*, *Mmp2*, and *Hif1a*. Selected examples are shown in Figure 2L through 2N over Slingshot Pseudotime. This analysis revealed that the proinflammatory module score modestly increases during EC phenotypic modulation in the same cells undergoing EndMT (Figure 2K; Figure S2E). This indicates that the inflammatory cell phenotype is not a distinct alternative EC phenotype, but rather one aspect of EC phenotypic modulation occurring simultaneously with partial EndMT.

In a more recent single-cell study, ECs have been described to acquire an immune cell-like phenotype (EndiCLT).^{1,41} However, on generating an EndiCLT module score based on the genes supporting this finding (*C1qa*, *C1qb*, *Tnf*, *C5ar1*, and *Lyz2*), we did not find evidence of this cell state in our aortic ECs (Figure S2F and S2G). Similarly, we did not detect a significant transition signal for the EndHT phenotype, defined by the markers *Kit*, *Sca1*, *Cd41*, *Cd45*, *Sox7*, *Sox17*, *Gata2*, *Runx1*, *Notch1*, *Bmp4*, *Tek*, and *Cd44*.^{1,42} (Figure S2H and S2I). Together, our data indicate that EC phenotypic modulation is best defined by EndMT and upregulation of a proinflammatory gene program, occurring together simultaneously in the same cells.

Figure 1 Continued. 10× scRNA-Seq. **E**, UMAP of all cells sequenced, grouped by broad cell types. **F**, A subset of ECs coexpressing lineage marker *tdT*, endothelial marker *Cdh5*, and arterial marker *Gja5*, highlighted in red. Cells are further grouped as Aortic ECs or microvascular (MV) ECs. **G**, UMAP of cells collected from the aortic arch after 0 weeks of HFD (orange) and after 16 weeks of HFD (purple) for all cells (**left**) or for *tdT*+/*Cdh5*+/*Gja5*+, aortic root ECs (**right**). **H**, UMAP of the aortic root ECs captured after 16 weeks of HFD, grouped by either quiescent or disease-exclusive modulated EC (ModEC) phenotypes. **I**, Heatmap of differentially expressed genes for quiescent (green bar) and ModECs (red bar). **J** through **L**, Expression of the quiescent aortic EC marker *Klk10* after 16 weeks of HFD, visualized by (**J**) FeaturePlot in the scRNA-seq data or by (**K** and **L**) RNAscope in situ hybridization within the distal arch. **M** through **O**, Expression of the ModEC aortic EC marker, *Edn1*, after 16 weeks of HFD, visualized by FeaturePlot (**M**) in the scRNA-seq data or by RNAscope (**N** and **O**) within the distal arch. All scale bars represent a length of 100 μm. SMC indicates smooth muscle cell; and Tmx, tamoxifen.

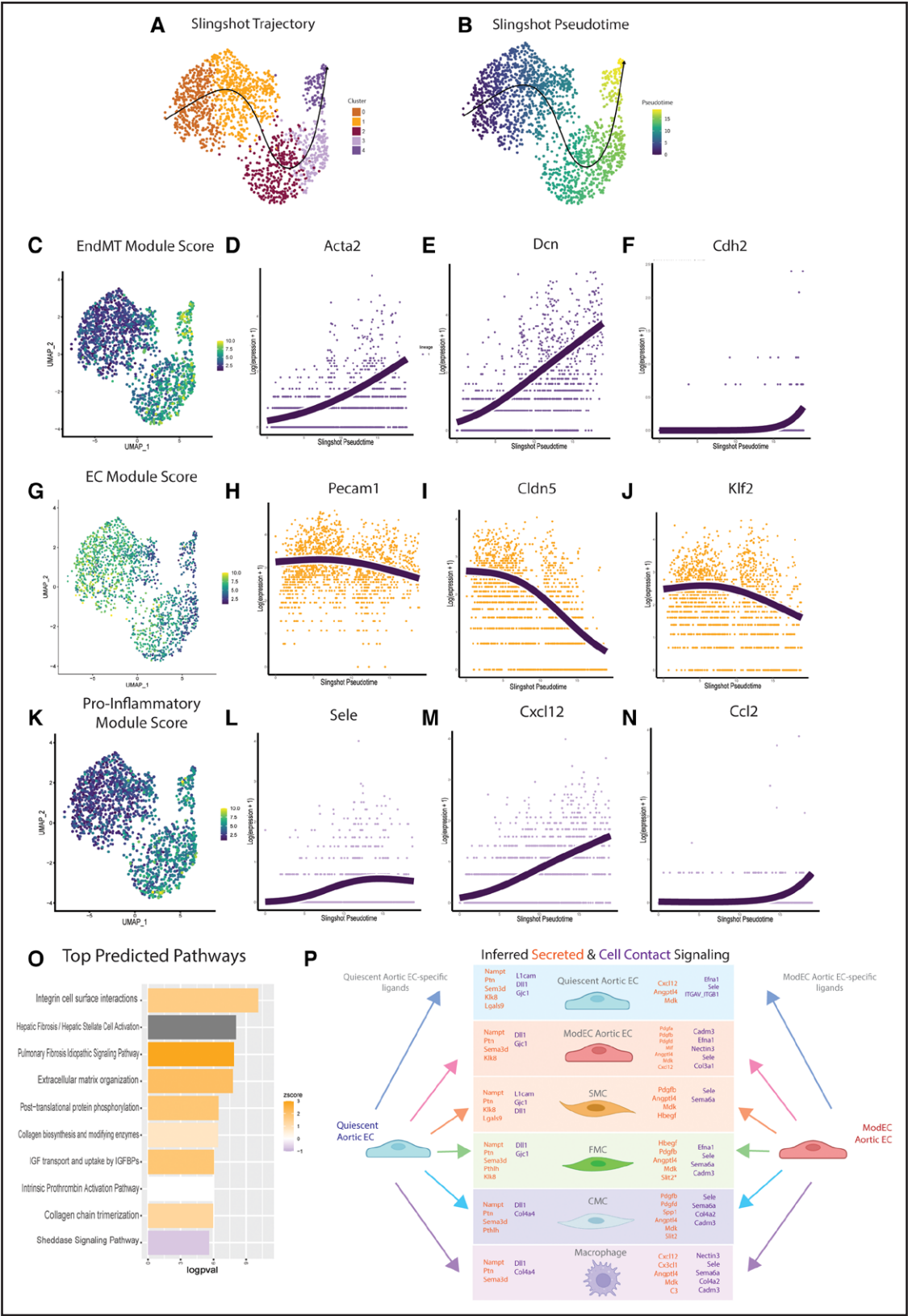


Figure 2. Characterization of endothelial cell (EC) phenotypic modulation and its predicted regulators during atherosclerosis. **A**, Slingshot trajectory on UMAP of aortic luminal ECs illustrates the predicted path through which ECs undergo phenotypic modulation. **B**, UMAP of aortic root ECs colored by Slingshot pseudotime. **C**, UMAP of the aortic root ECs colored by an endothelial-to-mesenchymal transition (EndMT) module score. **D** through **F**, TradeSeq smoothed plots for selected genes included in the EndMT module score, including *Acta2* (**D**), *Dcn* (**E**), and *Cdh2* (**F**) expression over Slingshot Pseudotime. **G**, UMAP of the aortic root ECs colored by an EC module (Continued)

Shifting to unbiased characterization of the gene programs unique to the disease-exclusive ModECs, pathway analysis of the DEG lists using Ingenuity Pathway Analysis³¹ showed upregulation of integrin cell surface interactions, fibrosis signaling, and ECM (extracellular matrix) organization in ModECs (Figure 2O). Top predicted upstream regulators were also identified (Figure S2J).

Interestingly, we observed distinct patterns of gene expression during EC phenotypic modulation. Although some genes, such as *Acta2*, *Dcn*, and *Cxcl12* (Figure 2D, 2E, and 2M), increased monotonically over pseudotime, others increased early in the ModEC transition and then plateaued (Figure 2L; Figure S2K). In contrast, other genes were only expressed at the most advanced stage of EC phenotypic modulation (Figure 2F and 2N; Figure S2L). This may suggest that, although the process of EC phenotypic modulation occurs across a continuum, distinct cell states may exist at different stages of phenotypic modulation (ie, early versus more advanced ModECs).

Cell-cell communication analysis (Figure 2P) identified key differences between quiescent and ModECs with respect to their predicted signaling to other cell types within the plaque. For example, the *Dll1* signaling to multiple cell types, present in quiescent ECs, is abolished in ModECs. In contrast, E-selectin (*Sele*) and several secreted signaling molecules such as *Cxcl12* and multiple *Pdgf* factors are upregulated in ModECs and predicted to interact with multiple cell types in the lesion. These results provide multiple mechanisms by which EC modulation might affect phenotypic changes in other cells during disease.

Comparative scRNA-seq Analysis of Mouse and Human Atherosclerotic Samples Reveals Conserved Endothelial Transition

To determine whether the EC phenotypic changes we observed in the mouse model of atherosclerosis are applicable to human atherosclerosis, we analyzed a scRNA-seq data set from coronary artery samples with no macroscopic atherosclerosis (nondiseased condition¹⁵), as well as human scRNA-Seq data from carotid endarterectomy samples as a diseased condition.¹⁶ *CDH5* was used to broadly identify the clusters of ECs within each data set (Figure 3A and 3C). Then, both *GJA5* (Figure 3B and 3D) and a human arterial EC module score (Figure S3A and S3B, see Methods) were used

to identify arterial ECs. The *CDH5*+/*GJA5*+ ECs from each human data set were integrated with the mouse *Twist1*-wt aortic root cells using the CONCORD algorithm¹⁷ (Figure 3E).

In the integrated data set, 2 markers of modulated ECs, *EDN1* and *NRG1*, are shown in Figure 3F and 3G, and a marker of quiescent ECs, *CLDN5*, is shown in Figure 3H. Interestingly, these markers roughly correspond to unbiased clustering at low resolution (Figure 3I), which likely represents quiescent and modulated ECs. Focusing exclusively on the human cells, we first tested whether the proportion of cells in the modulated EC cluster differed between the diseased and nondiseased samples. This analysis revealed that there was a greater proportion of modulated ECs in the diseased samples compared with the nondiseased samples (33% versus 15.6%, respectively, Figure 3J). Furthermore, expression of modulated EC-associated EC markers *EDN1* and *NRG1* was higher in ECs from the diseased samples compared with the nondiseased samples (Figure 3K and 3L). In contrast, we observed decreased expression of the quiescent-associated genes *CLDN5* and *KLF2* in the diseased samples (Figure S3C and S3D).

To determine whether the quiescent and disease-associated human ECs are indeed distinct populations when not coclustered with the mouse data, we visualized the human EC clusters from the carotid endarterectomy data set in the context of their native UMAP (Figure 3M and 3N). This demonstrated that the quiescent and modulated EC populations we identified do indeed have distinct transcriptional profiles in the human data set. In the carotid endarterectomy ECs, we next visualized the canonical EC module score based on expression of *CDH5*, *CLDN5*, *KLF2*, *VWF*, *PECAM1*, and *CD44* (Figure 3O). We also quantified the proinflammatory EC module score comprised of *ICAM1*, *SELP*, *SELE*, *CXCL12*, *MMP2*, and *HIF1A* (Figure 3P) and the EndMT module score comprised of *ACTA2*, *LUM*, *DCN*, *FN1*, *SNAI1*, *MMP2*, *TAGLN*, *TUBB3*, *TGFBR1*, *TGFBR2*, and *CDH2* (Figure 3Q). In addition, an EndHT module score comprised of *KIT*, *SOX7*, *SOX17*, *GATA2*, *RUNX1*, *NOTCH1*, *BMP4*, *TEK*, and *CD44* (Figure S3E) and an EndICLT module score comprised of *C1QA*, *C1QB*, *TNF*, *C5AR1*, and *LYZ2* (Figure S3F) were investigated. Altogether, the data revealed that the EndMT phenotype was most clearly recapitulated in the diseased human ECs (Figure 3O). The proinflammatory phenotype was also

Figure 2 Continued. score based on canonical EC markers. TradeSeq smoothed plots for selected genes included in the EC module score, including *Pecam1* (H), *Cldn5* (I), and *Klf2* (J) expression over Slingshot Pseudotime. K, UMAP of the aortic root ECs colored by a proinflammatory EC score. L through N, TradeSeq smoothed plots for selected genes included in the proinflammatory Module Score, including *Sele* (L), *Cxcl12* (M), and *Ccl2* (N) expression over Slingshot Pseudotime. O, Ingenuity Pathway Analysis (IPA) top biological pathways enriched in the differentially expressed gene list between modulated ECs (ModECs) and quiescent aortic ECs. The length of the bar represents $-\log_{10}(P \text{ value})$ assessed by Fisher exact test with Benjamini-Hochberg correction. Color of the bar represents the Z score computed from the IPA Ingenuity Knowledge Base. P, Overview of CellChat inferred cell-to-cell communications in *Twist1*-wt aortic arch cells, from source quiescent (left) or ModEC (right) aortic ECs to target cell types (middle). The ligands listed in orange represent the predicted secreted ligands, whereas the ligands listed in purple represent the predicted cell contact ligands. SMC indicates smooth muscle cell.

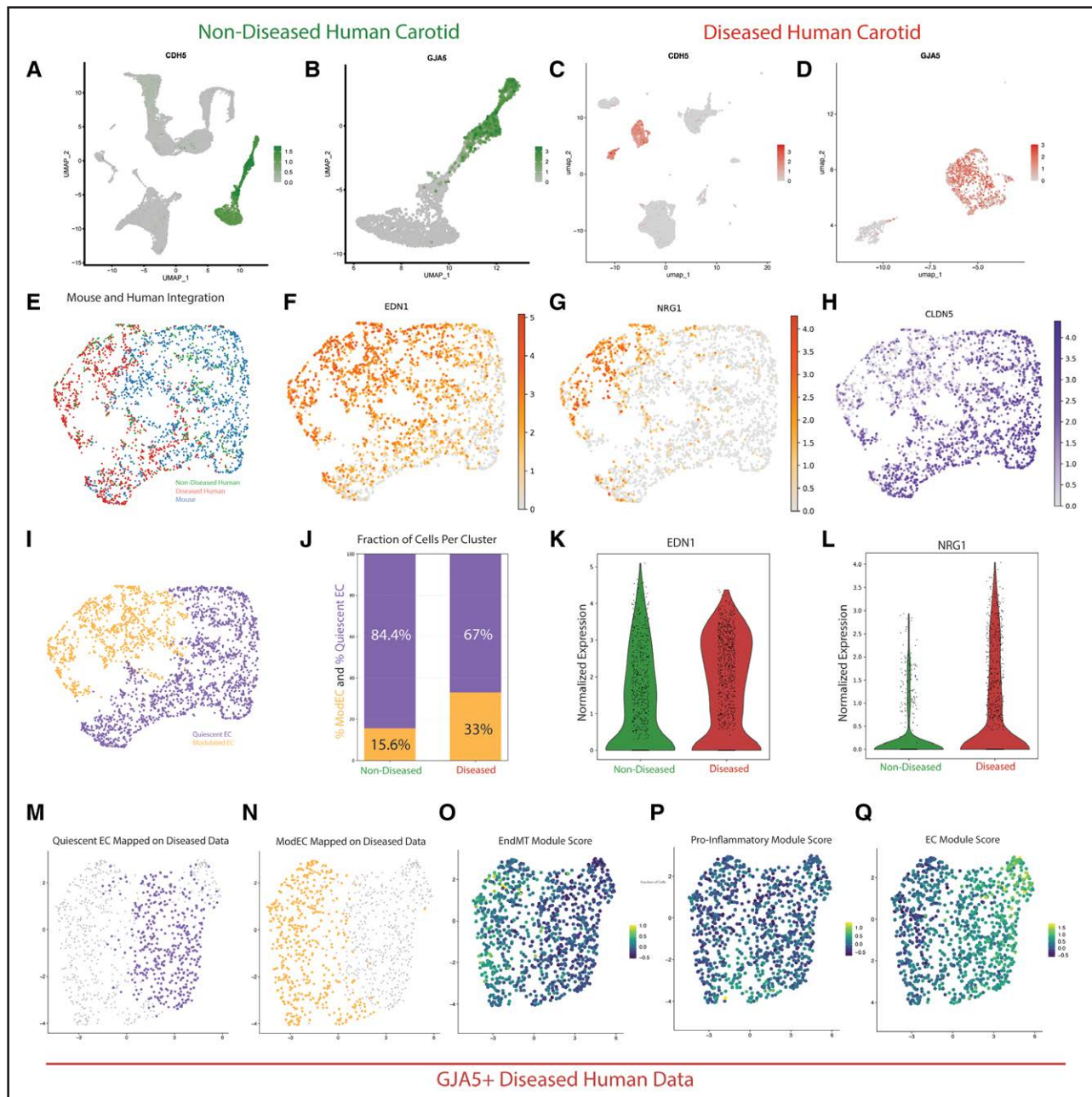


Figure 3. Comparative analysis of mouse and human endothelial cell (EC) phenotypic modulation.

A, UMAP of all cells sequenced from nondiseased human samples. CDH5+ ECs are highlighted in green. **B**, Subsetted UMAP of CDH5+ human ECs from nondiseased samples, with GJA5+ arterial ECs highlighted in green. **C**, UMAP of all diseased cells sequenced from human carotid endarterectomy samples. CDH5+ ECs are highlighted in red. **D**, Subsetted UMAP of CDH5+ human carotid endarterectomy ECs, with GJA5+ arterial ECs highlighted in red. **E**, UMAP of CONCORD-integrated human nondiseased (green), human diseased (red), and mouse aortic ECs (blue). **F**, Feature plot of EDN1 expression in all integrated cells. **G**, Feature plot of NRG1 expression in all integrated cells. **H**, Feature plot of CLDN5 expression in all integrated cells. **I**, Subsetted UMAP of all integrated cells, annotated as Quiescent EC or Modulated EC (ModEC). **J**, Bar plot representing the distribution of human ECs from diseased and nondiseased conditions in each cluster from **I**. **K** and **L**, Violin plot of EDN1 (**K**) or NRG1 (**L**) expression, grouped by nondiseased and diseased conditions. For **I** through **L**, proportions are descriptive; no statistical tests were performed. **M** and **N**, Native human UMAP of all diseased ECs, with cells assigned to Quiescent ECs in the joint CONCORD object colored in purple (**M**) or cells assigned to ModECs colored in orange (**N**). **O** through **Q**, UMAP of all diseased ECs with (**O**) endothelial-to-mesenchymal transition (EndMT) module score, (**P**) proinflammatory module score, or (**Q**) EC module scores plotted.

present, albeit to a lesser extent, in a subset of modulated ECs (Figure 3P). There was also a downregulation of canonical endothelial markers (Figure 3Q) in modulated

ECs. However, we did not observe an EndHT or EndICLT transition in the human data. To determine the pathways and regulators involved in EC phenotypic modulation in

human disease, we performed DEG analysis comparing the quiescent cluster (Figure 3M) to the modulated EC cluster (Figure 3N). We used Ingenuity Pathway Analysis to identify the top predicted pathways (Figure S3G), which were enriched with processes related to extracellular matrix production and consistent primarily with an EndMT phenotype. The top predicted upstream regulators of EC phenotypic modulation in human arterial ECs are shown in Figure S3H. Interestingly, some of these factors in the human data were also predicted to be drivers of EC phenotypic modulation in the mouse data (Figure S2I), such as the TEAD family of transcription factors and the MRTFA and MRTFB transcription factors. This is consistent again with a mesenchymal transition, and strongly implicates these factors in driving EndMT during atherosclerosis.

Twist1 Deletion Results in Decreased EC Phenotypic Modulation

To study the effect of *Twist1* deletion on EC phenotypic modulation and lesion characteristics during atherosclerosis, we crossed a conditional *Twist1* allele⁷ with the Twist1-wt line to generate an inducible *Twist1* knockout line (*Cdh5^{CreERT2}*, *Twist1^(flx/flx)*, *ROSA26^(tdT)*, *ApoE^(-/-)*) or Twist1-ECKO (Figure 4A). All experimental animals used in the study were age-matched when administered Tamoxifen. The 16-week HFD treatment was also consistent for all cohorts included in the comparison of phenotypic differences between genotypes. We isolated aortic arches from Twist1-wt and Twist1-ECKO male and female animals from above the aortic root to the level of the diaphragm and performed en face ORO staining (Figure 4B). Quantification of ORO percent area (male n=12 Twist1-wt, n=11 Twist1-ECKO; female n=9 Twist1-wt, n=10 Twist1-ECKO) revealed that Twist1-ECKO mice have significantly reduced lesion area in the aortic arch (Welch test male $P<0.001$, female=0.0079, Figure 4C). Although lesion area was reduced in the whole aorta in males (Mann-Whitney *U* test $P=0.037$, Figure 4D), this was driven by the difference in the aortic arch, as we found no significant difference in lesion area in the descending aorta in either sex (Figure S4A; male Welch test $P=0.88$, female=0.57). Similarly, in transverse sections of the aortic root (male n=16 Twist1-wt, n=19 Twist1-ECKO; female n=11 Twist1-wt, n=13 Twist1-ECKO), there was no significant difference in lesion area between genotypes (Welch test male $P=0.32$, female=0.41, Figure S4B).

tdTomato-positive ECs were also quantified in transverse sections of the aortic root (Figure S4C, annotated as in Figure S1A). There was no significant difference in the number of tdTomato-positive cells at the EC monolayer (Welch test male $P=0.24$, female=0.67, Figure S4D; male n=16 Twist1-wt, Twist1-ECKO n=19; female n=11 Twist1-wt, Twist1-ECKO n=13). However,

male Twist1-ECKO mice contained significantly fewer tdTomato-positive cells in the lesion below the EC monolayer (Welch test $P=0.003$; Figure 4E, n=16 Twist1-wt, Twist1-ECKO n=18) and a slight decrease in tdTomato-positive ECs in the vessel as a whole (Mann-Whitney *U* test $P=0.036$, Figure S4F). This indicates a reduction in EC phenotypic modulation or reduced proliferation/migration of phenotypically modulated ECs into the lesion.

To characterize the effect of *Twist1* deletion in ECs, we performed scRNA-Seq on a total of 19 Twist1-ECKO animals (8 males, 11 females) after 16 weeks of HFD, and the data were analyzed jointly with the Twist1-wt cells. Notably, among the ECs collected after 16 weeks of HFD, *Twist1* expression was identified exclusively in the aortic EC and valve EC populations (Figure S4G). Endothelial deletion of *Twist1* was validated in the scRNA-Seq data, showing a $\approx 95\%$ decrease in *Twist1* expression in the Twist1-ECKO ECs (Figure S4H). To determine how EC *Twist1* deletion affects EC phenotype during disease, the distributions of WT and ECKO cells were visualized in UMAP space (Figure 4E) and across EC phenotypic modulation pseudotime (Figure 4F and 4G). This revealed a bimodal distribution of cells, with quiescent cells at earlier pseudotimes and modulated ECs at later pseudotimes. In contrast to WT cells, which exhibited a greater density at later pseudotimes, the density of ECKO cells was greater at earlier pseudotimes (Figure 4G). Thus, *Twist1* deletion shifted ECs earlier in pseudotime, resulting in significantly less EC phenotypic modulation during disease ($P=0.002$, 2-sample Kolmogorov-Smirnov test). In addition, among modulated ECs, *Twist1* deletion appeared to disproportionately reduce modulated ECs at earlier pseudotimes, with no differences observed at the most advanced stages of phenotypic modulation (Figure 4G).

To validate the scRNA-Seq data, we performed situ hybridization of transverse aortic root sections to quantify *Edn1*, a highly specific ModEC marker (Figure 4H) in the vessel (Figure 4I). Quantification of expression (male n=17 Twist1-wt, n=16 Twist1-ECKO; female n=9 Twist1-wt, n=7 Twist1-ECKO) revealed that *Edn1* is significantly reduced at both the EC monolayer (Mann-Whitney *U* test male $P=0.001$, female $P<0.001$; Figure 4J) and in the lesion below the monolayer (Mann-Whitney *U* test male $P<0.001$, female $P=0.009$; Figure 4K) of Twist1-ECKO mice relative to Twist1-wt controls. Together, these data indicate that *Twist1* deletion in ECs indeed results in reduced EC phenotypic modulation, and that this reduction in EC phenotypic modulation occurs independently of migration into the lesion.

Surprisingly, our scRNA-Seq data revealed that in addition to reduced expression in ECs (reduced by approx. 95%), Twist1-ECKO mice also exhibited reduced *Twist1* expression in SMCs ($\approx 83\%$) and adventitial fibroblasts (AdvFibroblasts; $\approx 33\%$; Figure S4I). *Twist1* was virtually undetectable in macrophages in both genotypes

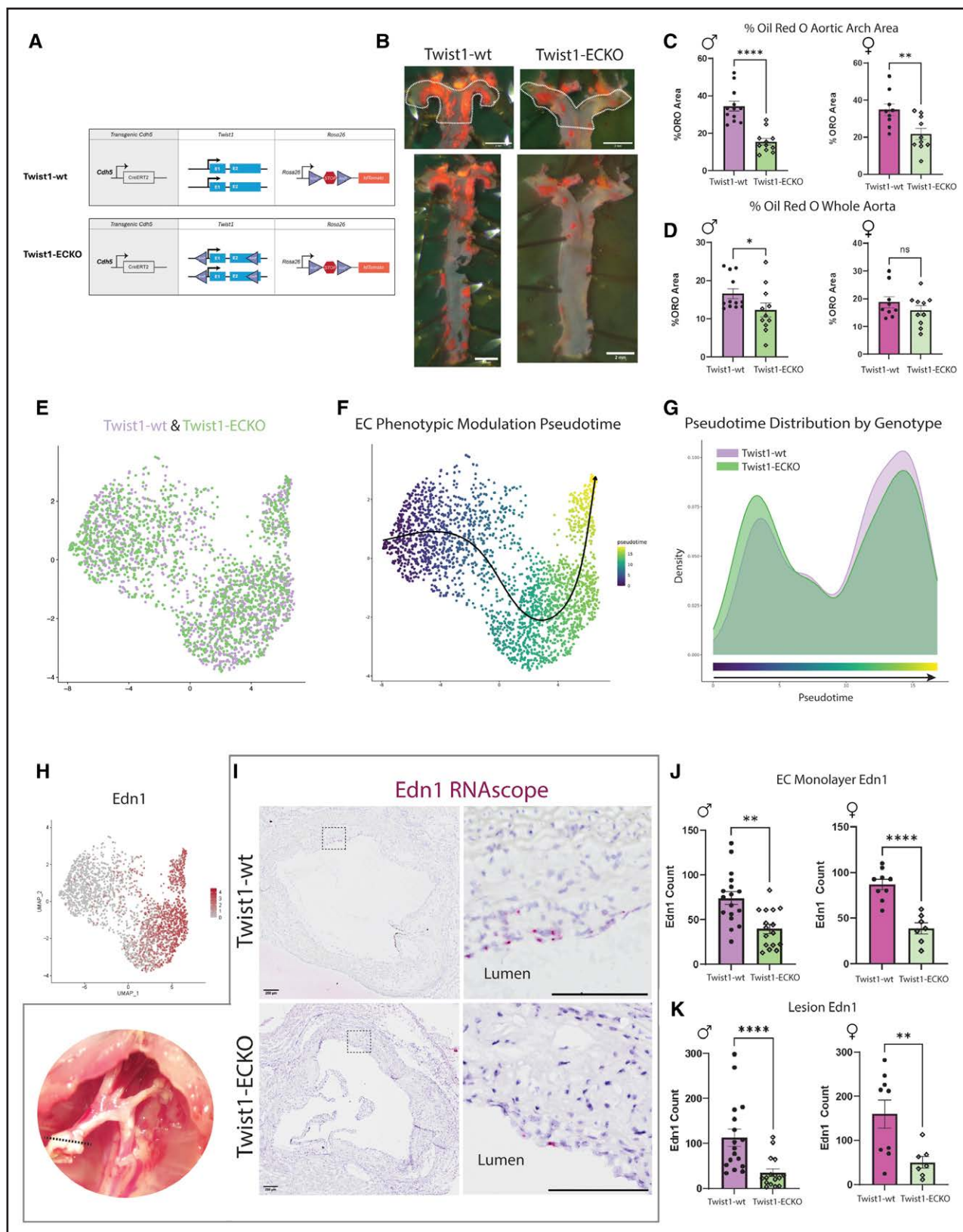


Figure 4. Endothelial cell (EC)-*Twist1* deletion inhibits EC phenotypic modulation and reduces lesion burden.

A, Schematic of lineage-traced Cre-lox system for endothelial *Twist1* deletion. **B**, Representative en face Oil Red O (ORO)-stained aortae of *Twist1*-wt (left) and *Twist1*-ECKO (right) male mice after 16 weeks of high-fat diet (HFD). The aortic arch area quantified is outlined with the white dashed line. All scale bars represent a length of 2 mm. **C**, Quantification of lesion area based on ORO staining in the aortic arch and **(D)** whole aorta, grouped by genotype and separated by sex (male $n=12$ *Twist1*-wt, $n=11$ *Twist1*-ECKO; female $n=9$ *Twist1*-wt, $n=10$ *Twist1*-ECKO). **(Continued)**

(data not shown). To determine whether this was because *Twist1* was being deleted from the genome, we visualized the scRNA-seq reads in each cell type in relation to the 3' loxP site (Figure S4J). In ECs from Twist1-wt mice, scRNA-seq reads extended past the 3' loxP site to the end of Exon 2, indicating that the *Twist1* gene had not undergone excision from the genome. In ECs from Twist1-ECKO mice, the few existing reads terminated before the 3' loxP site, indicating effective deletion of the *Twist1* locus from the genome. In contrast, in SMCs and adventitial fibroblasts, although the *Twist1* transcript levels were significantly reduced in Twist1-ECKO mice, the *Twist1* transcript reads extended past the 3' loxP site. This indicates that the *Twist1* locus remained intact within the genome, and suggests that *Twist1* expression is reduced in these mice by a mechanism other than Cre-mediated *Twist1* deletion. To confirm that reduced expression of *Twist1* in non-EC cell types was an intrinsic property of the *Twist1^{flx}* allele and not a Cre-mediated effect, we isolated RNA from aortic tissue extending from the aortic root through the distal arch from adult mice (Twist1-wt, Twist1-ECKO, and Twist1-flx genotypes) that did not receive any tamoxifen and assayed *Twist1* expression by qPCR. We found that the presence of the floxed *Twist1* allele alone significantly reduces *Twist1* expression in the absence of tamoxifen (Twist1-ECKO versus Twist1-wt, 1-way ANOVA, Tukey post hoc test $P=0.006$) and in the absence of both tamoxifen and Cre (Twist1-ECKO versus Twist1-flx, 1-way ANOVA, Tukey post hoc $P=0.004$, Figure S4K).

Endothelial-Specific *Twist1* Deletion Results in Reduced Lesion Size and Increased Lesion Stability in the Distal Arch

While our original model comparing Twist1-wt and Twist1-ECKO was necessary for lineage tracing of EC-derived cells, this led to reduced *Twist1* expression in multiple cell types and confounded investigation of EC-specific effects of *Twist1* on lesion phenotype. We therefore generated a cohort of mice that were homozygous for the *Twist1^{flx}* allele, with littermates either inheriting the Cdh5-Cre (Twist1-ECKO) or no Cre (Twist1-flx, Figure 5A). Because our ORO data showed a significant reduction in lesion size within the aortic arch, we characterized lesions within the distal arch adjacent to the left subclavian artery.

Transverse sections of the distal arch were prepared with Masson's Trichrome staining (Figure 5B). We observed a significant reduction in lesion area in Twist1-ECKO mice compared with Twist1-flx controls (males Twist1-flx $n=9$, $n=10$ Twist1-ECKO, Welch $P=0.048$, Figure 5C). Importantly, lesions from Twist1-ECKO mice had significantly reduced acellular area as a percentage of the lesion size (Welch $P=0.005$, Figure 5D), and significantly increased collagen content within the lesion (Welch $P=0.047$; Figure 5E) compared with Twist1-flx controls. There was no significant difference in media area (Figure S5A) nor vessel area (Figure S5B) between genotypes.

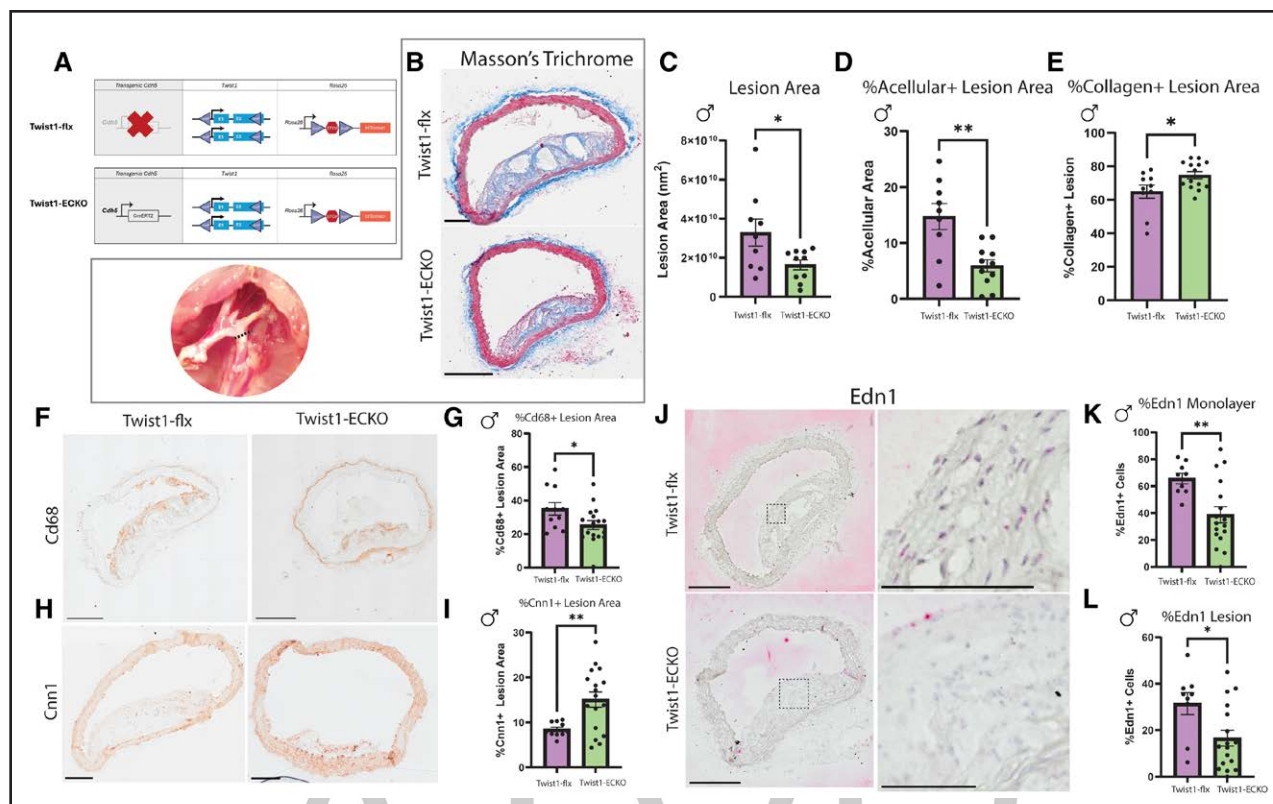
IHC staining for Cd68-positive macrophages (Figure 5F) revealed a significant reduction in Twist1-ECKO mice (males Twist1-flx $n=11$, Twist1-ECKO $n=17$, Welch $P=0.048$; Figure 5G). IHC staining was also performed for the SMC marker Cnn1 (Figure 5H), which revealed a significant increase in SMC content per lesion area in Twist1-ECKO mice (Welch $P=0.001$, Figure 5I). RNAscope situ hybridization was then performed to detect *Edn1* expression in the distal arch (Figure 5J). Quantification of *Edn1*-positive cells revealed a significant decrease at the monolayer (Twist1-flx $n=11$, Twist1-ECKO $n=17$ Welch, $P=0.020$; Figure 5K) and within the lesion below the monolayer (Mann-Whitney U test $P=0.017$, Figure 5L) in Twist1-ECKO animals. Sections were also stained with BODIPY 493/503 to assess neutral lipid content in the lesion (Figure S5C), which was found to be no different between genotypes (Figure S5D).

Taken together, these data strongly support that *Twist1* deletion inhibits endothelial phenotypic modulation in the lesion and at the distal arch, and that this results in smaller lesions with a more stable lesion phenotype. However, because the reduction in EC modulation mediated by *Twist1* deletion is not atheroprotective in all regions of the aorta, other interacting factors such as flow conditions are likely necessary for *Twist1* to affect lesion size and stability.

TWIST1 Regulates Downstream Targets in ECs Depending on Flow Conditions

To gain additional insight into the molecular mechanisms by which *TWIST1* alters EC biology, we transduced HCAECs with a *TWIST1* overexpression vector

Figure 4 Continued. Twist1-ECKO). Data in **C** were analyzed with a 2-tailed Welch t test and data in **(D)** were analyzed with a 2-tailed Mann-Whitney U test. **E**, Aortic root ECs with Twist1-wt cells displayed in purple and Twist1-ECKO cells displayed in green. **F**, UMAP of Twist1-wt and Twist1-ECKO aortic root ECs plotted over Slingshot Pseudotime. **G**, Distribution of Twist1-wt (purple) and Twist1-ECKO (green) ECs over Slingshot Pseudotime. Data were analyzed with a 2-sample Kolmogorov-Smirnov test. **H**, Feature plot of aortic *Edn1* expression. **I**, RNAscope situ hybridization for *Edn1* in transverse sections of the aortic root (dashed black line). Representative sections of Twist1-wt (**top**) and Twist1-ECKO (**bottom**) are shown. All scale bars represent a length of 250 μm . **J** and **K**, Quantification of *Edn1* RNAscope staining in the EC monolayer (**J**) or below the EC monolayer (**K**) of mice after 16 weeks of HFD, grouped by genotype and separated by sex (male $n=17$ Twist1-wt, $n=16$ Twist1-ECKO; female $n=9$ Twist1-wt, $n=7$ Twist1-ECKO). Data in **J** through **K** were analyzed with a 2-tailed Mann-Whitney U test. All data are presented as mean $\pm$ SEM.



or an EV control and placed cells on an orbital shaker for 5 days (Figure 6A). Cells at the center of each well, exposed to OSS disturbed-flow conditions, were isolated for RNA extraction and prepared for bulk RNA sequencing. DEGs induced by *TWIST1* overexpression were analyzed with Ingenuity Pathway Analysis for top predicted pathways, which included agranulocyte adhesion and diapedesis, RHOGDI signaling, and pathogen-induced cytokine storm signaling pathway (Figure 6B). In addition, *TWIST1* was identified as the top predicted upregulated upstream regulator, along with other factors with previously described roles in endothelial biology and pathobiology^{43–52} (Figure 6C).

In addition to investigating the effects of *TWIST1*-overexpression under proatherogenic OSS conditions,⁵³ transduced HCAECs were collected from the periphery of each well, as these cells were exposed to atheroprotective LSS⁵⁴ (Figure S6A). In contrast to the OSS-exposed transduced HCAECs, the top predicted pathways for LSS-exposed cells did not include

upregulated pathways with clear links to atherosclerosis (Figure S6B). Instead, upstream regulator analysis predicted suppression of inflammatory factors, such as TNF, NFKB1, NONO, IRF3, and IRF1, with *TWIST1* overexpression (Figure S6C).

When comparing DEGs with *TWIST1*-overexpression to EV controls separately according to LSS or OSS conditions, we identified 303 DEGs specific to the LSS condition, 229 DEGs specific to the OSS condition, and 402 DEGs shared between both flow conditions (Figure S6D). Analysis of top predicted pathways exclusive to either flow condition revealed downregulated inflammatory pathways in the LSS-only genes (Figure S6E), whereas atherosclerosis-relevant and extracellular matrix pathways were retained in the OSS-only genes (Figure S6F). Together, our data strongly suggest that the effect of *TWIST1* varies significantly with flow conditions in ECs, and that disturbed-flow conditions are important for *TWIST1* to regulate atherosclerosis-relevant pathways in ECs.

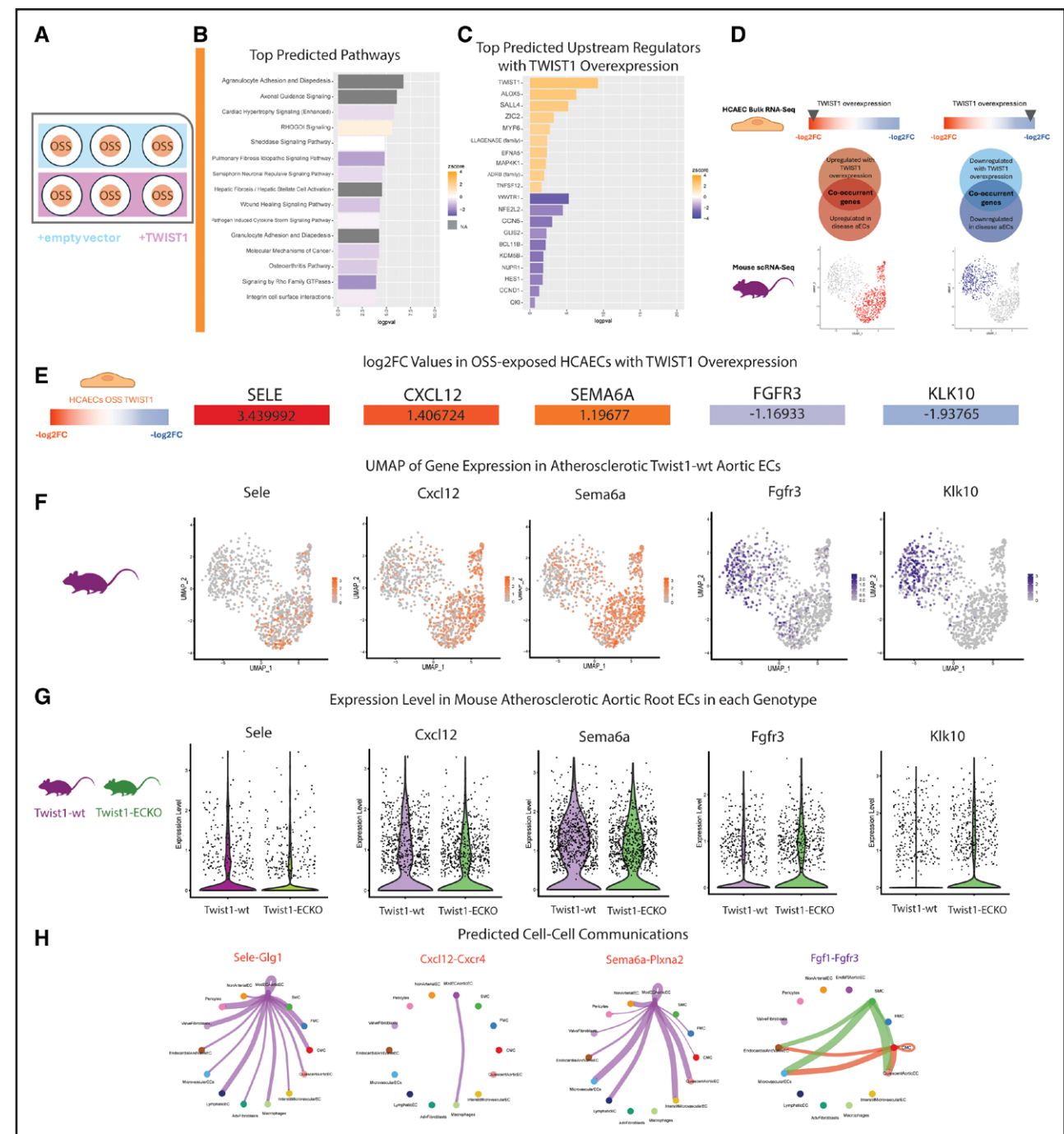


Figure 6. Identification of *Twist1*-specific mechanisms during flow and disease states.

A, Schematic of 6-well plate orbital shaker flow model, with oscillatory shear stress (OSS)–exposed human coronary artery endothelial cells (HCAECs) at the center of each well. Cells were transduced with lentivirus containing either an empty vector (EV) or a TWIST1-overexpression construct. **B**, Ingenuity Pathway Analysis (IPA) top biological pathways enriched in the differentially expressed gene list between HCAECs with TWIST1 overexpression compared with EV-treated cells, under OSS conditions. Gray bars indicate no Z score value predicted. **C**, IPA top predicted upstream regulators of OSS-exposed HCAECs treated with a TWIST1 overexpression vector as compared with the EV control. For **B** and **C**, length of the bar represents $-\log_{10}(P \text{ value})$ assessed by Fisher exact test with Benjamini-Hochberg correction. Color of the bar represents Z score computed from the IPA Ingenuity Knowledge Base. **D**, Schematic of the analysis performed to identify the subset of co-occurrent genes involved in endothelial cell (EC) phenotypic modulation that are regulated by TWIST1. **E**, $\log_2\text{FC}$ values of selected co-occurrent genes *SELE*, *CXCL12*, *SEMA6A*, *FGFR3*, and *KLK10* in the HCAEC bulk RNA-Seq data. **F**, Feature plots of the same co-occurrent genes in the luminal aortic ECs (aECs) of the Twist1-wt mouse scRNA-Seq data after 16 weeks of high-fat diet (HFD). **G**, Violin plots of the co-occurrent gene expression levels in Twist1-wt and Twist1-ECKO mouse scRNA-Seq data after 16 weeks of HFD. **H**, Predicted cell-cell signaling pairs in Twist1-wt mice after 16 weeks of HFD for the selected co-occurrent genes. *Klk10* was not present in the cell-cell communication database. ModEC indicates modulated EC; and SMC, smooth muscle cell.

***TWIST1* Regulates a Subset of Genes Involved in EC Phenotypic Modulation**

To more clearly identify which genes in vivo in EC modulation are specifically regulated by *TWIST1*, we queried both the mouse aortic EC scRNA-Seq data from the 16-week HFD time point and the *TWIST1* overexpression HCAEC Bulk RNA-Seq data (Figure 6D). Because *TWIST1* promotes EC modulation, we sought to identify 2 groups of genes: (1) genes that were both activated by *TWIST1* overexpression in HCAECs and activated during EC modulation in disease; and (2) genes that were both suppressed by *TWIST1* overexpression in HCAECs and suppressed during EC modulation in disease. Together, these lists comprised 26 genes that *TWIST1* may regulate during EC phenotypic modulation (Table S1). Several upregulated genes have clear functions related to EC phenotypic modulation, including TGF (transforming growth factor)- β and FGF-2 (fibroblast growth factor-2) signaling and immune cell chemotaxis/adherence. Genes related to semaphorin-plexin signaling were also coordinately downregulated. Selected genes are displayed in Figure 6E and 6F. When stratified by mouse genotype, genes that were upregulated by *TWIST1* overexpression in HCAECs were downregulated in *Twist1*-ECKO mice, and those downregulated by *TWIST1* overexpression in HCAECs were upregulated in *Twist1*-ECKO mice (Figure 6G; Table S1). As these genes are regulated in opposite directions with *TWIST1* overexpression versus *TWIST1* deletion, they represent a higher-confidence *TWIST1* regulatory program in ECs during atherosclerosis.

To determine how these genes might affect the multicellular plaque environment, we identified potential secreted and cell-cell contact communications using CellChat in the larger scRNA-Seq data set containing all cell types (Figure 6H). *Fgfr3* was predicted to only receive *Fgf1* signaling on quiescent aortic ECs, which is consistent with observations from previous studies showing that FGF signaling promotes vascular resilience and prevents EndMT.⁵⁵ In contrast, *Cxcl12* and *Sele* are predicted to signal from ModECs to other cell types in the plaque, particularly macrophages. Indeed, the induction of CXCL12 and E-selectin by *Twist1* in ModECs is a potential molecular mechanism underlying our observation that fewer macrophages are present in the distal arch lesions of *Twist1*-ECKO mice.

DISCUSSION

The response of ECs to atherogenic stimuli has been studied extensively in vitro, leading to fundamental insights regarding the potential role of these cells during disease. The study of EC-specific processes in vivo has been more challenging due to the multicellular composition of the atherosclerotic plaque. This has improved

with the use of lineage tracing, but the traditional assessment of only a small number of markers has been a fundamental barrier to the comprehensive assessment of EC phenotype. More recently, with the ability to assay thousands of genes simultaneously in individual cells, single-cell omics have the promise both to reveal new aspects of EC biology and to place previous observations in context. However, as most current techniques require dissociation of the tissue, spatial information is lost. As a result, bystander ECs residing in the surrounding structures, including the vascular adventitia, valve apparatus, or adjacent tissues, can be misinterpreted as EC heterogeneity at the vessel lumen. This can result in erroneous conclusions regarding ECs transitioning between clusters, when in reality, the clusters represent unrelated ECs in different locations. In the present study, we leveraged (1) established markers of arterial EC identity; (2) our previous work localizing EC subtypes within the aortic root¹⁴; and (3) additional EC subtype localization to identify the EC population at the luminal surface that directly participates in the atherosclerotic lesion. Importantly, this was the only EC population to undergo a significant phenotypic shift between baseline and 16 weeks of HFD, reinforcing this conclusion.

We found that these cells appear to undergo a single-cell fate transition, which encompasses both proinflammatory and EndMT phenotypes. Thus, rather than being 2 alternative phenotypes, these attributes occur simultaneously in the same cells during EC phenotypic modulation. Notably, we did not find evidence of ECs undergoing transition to immune-like (EndICLT) or hematopoietic-like (EndHT) cells in our model system. This contrasts with a recent report¹ that used scRNA-seq and scATACseq in a carotid model of disturbed flow and atherosclerosis. This discrepancy is likely due to the different models used between the studies. Overall, it is vital that only ECs that are located at the monolayer overlying the lesion or within the lesion are included in the analysis. Inclusion of bystander ECs in other anatomic locations (valvular ECs, adventitial microvascular ECs, and lymphatic ECs) in cell fate trajectory analyses can interfere with interpretations of such cellular transitions. More broadly, we did not identify multiple discrete EC fate trajectories in mouse or human scRNA-seq data, but this possibility is difficult to rule out given current technologies. Our analysis also revealed evidence of distinct pseudotemporal gene expression patterns (Figure 2D through 2F, 2H through 2J, and 2K through 2M; Figure S2L through S2N), which may indicate that a distinct phenotype exists at later pseudotime values within the ModEC population. Interestingly, ECs that demonstrated the most advanced degrees of phenotypic modulation (later pseudotime values, and marked by expression of *Nrg1*) were found at the monolayer, as well as within the lesion, suggesting that EC migration may be uncoupled from dedifferentiation and acquisition of a mesenchymal/inflammatory phenotype.

In contrast, it is likely that proliferation and migration are tightly coupled during EC phenotypic modulation, as we only observed increased numbers of ECs where cells had migrated into the lesion, and not at the monolayer. Future studies using spatial transcriptomics may be able to identify factors common to ECs that proliferate and migrate into the lesion versus those that remain at the monolayer. Importantly, the human data analyzed in our study mirrors many of the same phenotypic changes we observed in lineage-traced mouse ECs, primarily with respect to the EndMT phenotype. The human ECs showed upregulation of genes in the proinflammatory module score to a lesser extent than the mouse ECs. It is possible that an initial inflammatory EC phenotype is reduced over time, with the EndMT phenotype predominating at the time of human tissue collection.

The CAD risk allele at 7p21.1 is associated with increased *TWIST1* expression in vascular tissue in the GTEx data set, indicating that *TWIST1* expression promotes lesion formation and risk for myocardial infarction. This is directionally consistent with our current findings that EC-specific *Twist1* deletion results in reduced lesion burden and a more stable plaque phenotype. Our findings contrast with a recent study⁵ reporting a lower lesion burden but a more stable lesion phenotype with *Twist1* deletion in ECs. This discrepancy could be due to differences in experimental design. For example, the investigators induced atherosclerosis by giving HFD for 8 weeks before deleting *Twist1* in ECs, followed by another 8 weeks of HFD, whereas we deleted *Twist1* before the onset of atherosclerosis. Our study may be more consistent with human genetic data because it more closely mimics the long-term difference in *TWIST1* expression mediated by an inherited allele. In contrast, it is possible that reduction in EC *Twist1* has different consequences when induced later during disease, which may have implications for potential therapies targeting *TWIST1*. Another major difference is our discovery that the *Twist1^{flx}* allele is hypomorphic, resulting in reduced basal *Twist1* expression in multiple cell types even without Cre-mediated excision of the *Twist1* gene. Because plaque composition is influenced by multiple cell types, altered *Twist1* expression in multiple cell types fundamentally confounds any interpretation of the role of *Twist1* specifically in ECs. We therefore generated a second cohort of mice (*Twist1^{flx/flx}; +/-Cre*) which allowed us to cleanly determine the effect of EC-specific *Twist1* deletion on plaque architecture. Thus, our study also differs in the genetic composition of mice studied.

Although the hypomorphic *Twist1^{flx}* allele confounds interpretation of plaque phenotype in our first (lineage tracing) cohort of mice, cell-autonomous effects of *Twist1* in ECs, such as the effect of *Twist1* on EC phenotypic modulation, should be preserved. Indeed, we observed that deletion of *Twist1* in ECs in both cohorts of mice markedly reduced EC phenotypic modulation,

as evidenced by reduced *Edn1* expression at the EC monolayer and within the plaque. In addition to promoting EC phenotypic modulation, it is also likely that *Twist1* affects both EC proliferation and migration based on the changes in EC numbers we observed in the lesion and monolayer in our first lineage tracing cohort of mice (*Twist1*-ECKO versus *Twist1*-wt).

Integration of bulk RNA-seq data of OSS-exposed, *TWIST1*-overexpressing HCAECs with the scRNA-seq data of mouse aortic ECs provided more insight into the regulatory effects of *TWIST1*. As examples, our data collectively show that *TWIST1* upregulates *CXCL12* and *SELE*. Importantly, *CXCL12* is the causal gene at the 10q11 CAD risk locus,⁵⁶ and a recent study identified ECs as the relevant source of *CXCL12* during atherosclerosis, with EC-specific *CXCL12* deletion resulting in decreased *CXCL12* plasma levels and reduced lesion area in the aorta.⁵⁷ Our study directly links these 2 important CAD GWAS signals, suggesting that this is a critical nexus of CAD risk mediated through ECs. Another *TWIST1* target, *SELE*, is synthesized during inflammation by activated ECs^{58,59} and found on the surface of fibrous and lipid-containing plaques^{58,60} where it likely mediates monocyte adhesion during atherosclerosis. Notably, when comparing OSS versus LSS conditions, *SELE* was differentially regulated only in the setting of *TWIST1* overexpression, and not in cells transduced with EV. This suggests that both *TWIST1* and disturbed-flow conditions are required for *SELE* upregulation, which may provide a mechanism underlying the finding that *Twist1* deletion reduces disease only in specific regions of the aorta.

In contrast, our data collectively show that *TWIST1* downregulates *KLK10* and *FGFR3* expression. With regard to *FGFR3*, loss of FGF signaling has been reported to promote EndMT, promote deposition of fibronectin, and increase neointima formation in a mouse model of atherosclerosis,⁶¹ possibly via upregulation of TGF- β signaling.^{55,62}

Together, our data demonstrate that *TWIST1* affects risk for atherosclerosis via ECs by promoting proliferation, migration, and phenotypic modulation, ultimately leading to an unstable plaque phenotype. Importantly, we have also identified specific mechanisms in ECs by which *TWIST1* likely promotes lesion progression. However, *Twist1*-induced EC phenotypic modulation does not appear to be sufficient to increase lesion burden, as we did not observe changes in lesion size in the aortic root or the descending aorta. Rather, as demonstrated by the change in disease burden and lesion stability in the aortic arch with EC *Twist1* deletion, and by the significant difference in gene expression programs between *TWIST1*-overexpressing cells in OSS versus LSS conditions, it is likely that the effect of *TWIST1* on disease progression depends strongly on the local flow conditions within the vessel. Finally, although we have provided evidence for

a causal role for *TWIST1* in ECs, it will be important to determine whether *TWIST1* is relevant in other cell types that may modify lesion size and architecture, and the resulting risk for vascular disease.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Table S1
Figures S1–S6
Major Resources Table
ARRIVE Checklist

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